Software Requirements Specification for Flexin'

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Revision History

Date	Version	Notes
October 19, 2026	1.0	Revision 0

1 Purpose of the Project

This section outlines the user business context and the overarching project goals for the Flexin' application.

1.1 User Business

Personal finance and physical fitness both depend on the principle of compounding, where consistent, early actions yield substantial long-term benefits; however, young adults often neglect both disciplines due to delayed gratification and widespread gaps in financial literacy. While existing software tools address workout tracking and stock market simulations independently, none connect daily physical activity directly to experiential investing. The business context of Flexin' centers on bridging this gap by using healthy, habit-building fitness routines as the gateway to practical financial education, as detailed in the Problem Statement and Goals and Development Plan.

1.2 Goals of the Project

The primary goal of Flexin' is to teach users the power of compounding by rewarding tracked physical activity with virtual currency that can be invested in a simulated financial market of fictional stocks, bonds, and exchange-traded funds (ETFs) whose prices follow real-world market trends. In addition, the project aims to integrate seamlessly with existing Android health technologies and wearable devices, deliver an intuitive user experience accessible to beginners, and drive sustained user engagement to cultivate lasting wellness and investing habits. Further details regarding project scheduling and project milestones are outlined in the Development Plan and Problem Statement and Goals.

1.3 About This Document

This Software Requirements Specification (SRS) describes what Flexin' must do, without prescribing how it is built. It is written for the development team, the project supervisor, teaching assistants and reviewers, and new team members who need to get up to speed on the project.

1.3.1 Template

This document follows the Volere Requirements Specification Template by Suzanne and James Robertson, as provided in the course's capTemplate repository. The following changes were made to the template:

- The Risks section was removed, because risks are covered by the Hazard Analysis.
- This subsection (About This Document) was added to Section 1 to record the template, the conventions, and the symbolic constants.
- Requirements are presented in tables with identifier (ID), Requirement, and Specification columns, rather than as individual requirement cards. Each specification has the Description, Rationale, and Fit Criterion fields of the Volere card, plus a Hazard Analysis trace and a list of related requirements where they apply.
- A Likely and Unlikely Changes section was added after Open Issues (Section 19).
- A traceability matrix was added as an appendix (Appendix 28), along with the course's Generative Artificial Intelligence (AI) Use Disclosure and Reflection appendices.

1.3.2 Conventions

- "Shall" marks a mandatory requirement. Requirements are written to be black-box: they say what the product must do, not how. Design decisions that are imposed on the team are recorded as solution constraints (SC-01 to SC-06).
- Every requirement has a unique ID. IDs use the prefixes in Table 1 followed by a two-digit number. Every mention of a requirement ID is a hyperlink to its definition, and the traceability matrix lists, for each requirement, every requirement it relates to in both directions.
- Values that appear in several requirements, or that the team will tune, are symbolic constants. They are written in sans-serif, link to Table 2, and are defined only there.

- References such as SR8 and H4.1 refer to a safety requirement (SR) and a hazard in the Hazard Analysis.
- Amounts of virtual currency use the virtual currency symbol (FLX). Days, dates, and times of day are in the user's local time zone.
- Acronyms are spelled out at first use and are listed in the Glossary (Section 4.1) with a link to where each is defined. Not applicable (N/A) marks template sections that do not apply to this project.

Prefix	Meaning and location
SC	Solution constraint (Mandated Constraints)
BE, BUC	Business event (BE) and business use case (BUC) (The Scope of the Work)
FUNC	Functional requirement
LF-APP, LF-STY	Look and feel: appearance, style
UHR-EOU, UHR-PER, UHR-INT, UHR-LRN, UHR-UND, UHR-POL, UHR-ACC	Usability and humanity: ease of use, personalization, internationalization, learning, understandability, politeness, accessibility
PR-SL, PR-SC, PR-PA, PR-RFT, PR-C, PR-SE, PR-L	Performance: speed and latency, safety-critical, precision and accuracy, robustness and fault tolerance, capacity, scalability and extensibility, longevity
PROD	Productization (Operational and Environmental Requirements)
MAS-MNT, MAS-SUP, MAS-ADPT	Maintainability and support: maintenance, supportability, adaptability
SEC-ACC, SEC-INT, SEC-PRV, SEC-AUD, SEC-IMM	Security: access, integrity, privacy, audit, immunity
CUL	Cultural requirement
COMP-LEG, COMP-STD	Compliance: legal, standards
DOC, TRN	User documentation, training
LC, UC	Likely change, unlikely change (Section 19)

Table 1: Requirement ID prefixes

1.3.3 Symbolic Constants

Table 2 defines the symbolic constants used in the requirements. Values marked "initial" are starting values that the team will confirm at the Currency Decision milestone (November 9, 2026) and adjust after user testing. Changing a value here changes it everywhere it is used.

Table 2: Symbolic constants

Constant	Value	Used in and basis
HealthyDailyActivity	10,000 steps or 60 minutes of recorded workouts (initial)	PR-SC-01. A widely used daily activity target.
DailyEarnCap	The currency earned for 1.5 × HealthyDailyActivity (initial)	PR-SC-01. Removes any reward for activity far beyond a healthy volume.
MaxGoalIncrease	10% per week (initial)	PR-SC-06. Follows the common guideline of increasing training load by no more than 10% a week.
MissedGoalLimit	3 consecutive goal periods (initial)	PR-SC-06.
MaxDailyNotifications	3 per day (initial)	PR-SC-07.
OfflineSyncWindow	7 days	PR-RFT-01. Covers a week away from a connection or without opening the app.
AccessExpiry	60 minutes	SEC-ACC-01.
SessionTimeout	30 days without use	SEC-ACC-05.
MinPasswordLength	8 characters	SEC-ACC-04.
LockoutAttempts	5 consecutive failures	SEC-ACC-04.
LockoutDuration	15 minutes	SEC-ACC-04.
MaxCadence	250 steps per minute, sustained	SEC-INT-01. Faster than a person can sustain, so it indicates faked data.
RateLimit	60 requests per minute	SEC-IMM-02.
BackupInterval	24 hours	SEC-INT-03.
BackupRetention	30 days	SEC-INT-03, SEC-PRV-03.

Constant	Value	Used in and basis
RestoreTime	24 hours	SEC-INT-03.
DeletionWindow	24 hours	SEC-PRV-03.
MinAge	18 years	COMP-LEG-06. Age of majority for consenting to share personal health information.

1.3.4 Diagram Notation

The context diagram (Figure 1) shows the Flexin' system as a rounded box and each external entity as a rectangle; each double-headed arrow is information exchanged in both directions and is labelled with that information. The business data model (Figure 2) is a Unified Modeling Language (UML) class diagram. The product use case flowcharts in Section 8.3 use the notation described at the start of that section.

2 Stakeholders

This section will look at all individuals and groups that will be affected by the success of this project. We will look further and study the people affected through the following sections.

2.1 Client

The client with whom the team is building the app is **Chris Schankula**. Chris was responsible for providing the initial idea and will work throughout each stage of the project to deliver the final product.

2.2 Customer

- **Primary**: Young adults aged 18-30 who are seeking to build financial literacy and sustainable fitness habits.
- **Secondary:** Adults older than the intended age range who are looking to learn new skills, reinforce or start new healthy habits.

2.3 Other Stakeholders

- **Financial Institution/Banks:** which benefit from financially literate consumers who will engage more with their finances.
- **Gyms/Training Facilities:** which can benefit from increased user activity by providing a place to exercise.

2.4 Hands-On Users of the Project

- **Mobile (Android) Smartphone Users:** users will be able to interact with the app and utilize the different features to view and track financial and fitness goals. For the purpose of Capstone our final deployed app will be for Android users only.

The hands-on users have the following characteristics. The personas below illustrate them.

- **Age:** adults (MinAge or older), mostly 18 to 30, including university students and people early in their careers.
- **Financial knowledge:** most are beginners. Among Canadians under 35, only about 30% could answer three basic questions on compound interest, inflation, and diversification (**?**). A smaller group, such as the Connor persona, already knows a lot about investing and wants depth and challenge.
- **Fitness level:** ranges from inactive to regular athletes. Some users have injuries, disabilities, or health conditions that limit the activities they can do.
- **Technology:** own an Android phone and are comfortable with mobile apps; some also own a Garmin watch. They are not expected to have any technical knowledge beyond everyday app use.
- **Usage pattern:** short sessions on the go, often right after a walk or workout, sometimes with a poor connection.
- **Language and location:** live in Canada and use the app in English or French.

- **Accessibility:** some users have visual, hearing, or motor impairments, or colour vision deficiency.
- **Motivation:** respond to goals, streaks, and friendly competition, but can be discouraged by repeated failure or by comparisons with others.

2.5 Personas

2.5.1 Brandon - The Novice Saver and Avid Runner

Background

Brandon is a university student in his final year of Environmental Sciences at the University of Toronto, preparing for graduation in the spring. Brandon has recently begun learning more about his own finances, but still has a limited understanding of investing and saving.

Goals and Motivations

Brandon enjoys recreational running as a way to stay active, relieve stress, and maintain a healthy lifestyle. He wants to improve running performance, build consistency, and take this hobby to the next level. At the same time, Brandon wants to develop his financial literacy and gain confidence in investment decisions before investing real savings.

Behaviours and Habits

Brandon balances school responsibilities with running and personal development. He currently uses his phone as his main device to track his running activities such as goals and progress. Brandon prefers simple explanations, interactive learning, and measurable progress rather than complicated financial terminology.

Needs and Expectations

- A risk-free stock trading simulator that uses virtual money.
- Beginner-friendly lessons explaining investment opportunities, market trends, risk, and investment strategies.

- Progress tracking and feedback to help build investing knowledge and confidence.
- Running features to record workouts, monitor distance and pace, and track improvements over time.
- Achievable goals, milestones, and challenges that encourage consistent participation in both activities.

Frustrations

Brandon finds financial platforms intimidating because of unfamiliar terminology and the potential to lose money. Many investing resources assume prior knowledge, while running apps may focus primarily on recording workouts rather than helping beginners improve. Brandon wants a straightforward experience that makes learning and progress rewarding.

2.5.2 Connor - The Young Expert

Background

Connor is a third-year Commerce student specializing in Finance, returning to university after completing a year-long co-op placement at IG Wealth Management. Coming from a long line of financial professionals, Connor has developed a strong interest in financial management and aspires to build a successful career in the industry. As a top-performing student and hard-working individual, Connor is confident in his financial knowledge and enjoys applying what he has learned to real-world situations.

Goals and Motivations

Connor wants to excel academically and build on the professional experience gained during his co-op placement to advance his career in financial management. He enjoys demonstrating his financial knowledge and sharing his opinions on investment opportunities and market trends. Although Connor is an avid sports fan, he recognizes that balancing academics, career ambitions, and personal interests leaves him with little time to exercise. He wants to find ways to incorporate physical activity into his busy schedule without compromising his professional goals.

Behaviours and Habits

Connor maintains a demanding schedule focused on academics, career development, and expanding his financial knowledge. He regularly follows financial news, monitors market trends, and enjoys discussing investment strategies with others. Connor is competitive and motivated by measurable results, academic achievement, and opportunities to demonstrate his expertise. While he enjoys watching sports, exercise is not currently a consistent part of his routine due to his time commitments.

Needs and Expectations

- Advanced financial tools and simulated trading features that allow him to apply his existing investment knowledge.
- Access to market trends, investment opportunities, and performance statistics to support informed decision-making.
- Competitive features such as rankings, challenges, and performance comparisons to showcase his financial expertise.
- Efficient running and exercise tracking that fits into his busy academic and professional schedule.
- Personalized fitness goals, progress tracking, and achievable challenges that encourage consistent physical activity.
- An engaging platform that combines financial competition with physical activity, allowing him to pursue both interests in one place.

Frustrations

Connor finds it difficult to balance his academic workload, career ambitions, and personal interests while maintaining an active lifestyle. Beginner-focused financial platforms may not provide enough depth or challenge to keep him engaged, while traditional fitness applications may not offer the competitive elements that motivate him. Connor wants an efficient, challenging experience that allows him to demonstrate his financial knowledge while encouraging him to make time for exercise.

2.5.3 Daniella — The All-Arounder

Background

Daniella is a recent Civil Engineering graduate from the University of Guelph who gained valuable work experience and earned a substantial amount of money through multiple co-op placements throughout her degree. Through personal curiosity and working with her bank to understand her finances, Daniella developed a strong awareness of saving and managing money. Following graduation, she dedicated significant time to building her own investment portfolio and developing her financial knowledge. Although she has gained practical experience managing her savings, Daniella wants to become more confident in making independent investment decisions.

Goals and Motivations

Daniella enjoys staying active through a variety of organized sports which she participates in throughout the year. She worries that the time commitment required by multiple sports leagues may eventually become unsustainable, and makes the decision to cut back on some of the organized leagues that she participates in. She is interested in beginning a more traditional circuit-style workout as these tend to be more convenient. Financially, she wants to continue growing her investment portfolio while gaining more experience evaluating opportunities and making important investment decisions independently.

Behaviours and Habits

Daniella balances her professional responsibilities and personal interests while maintaining an active social life through organized sports. She prefers physical activities that are engaging, competitive, and social. Financially, Daniella takes an active role in monitoring her savings and managing her investment portfolio. She is curious, self-motivated, and willing to learn independently, but recognizes that managing a portfolio with real money offers limited opportunities to experiment with unfamiliar strategies without risking her savings.

Needs and Expectations

- A realistic stock trading simulator that allows her to practice making independent investment decisions without risking real money.
- Investment tools that help evaluate market trends, compare opportunities, and understand the potential risks and outcomes of different strategies.
- Performance tracking and feedback to evaluate investment decisions and identify areas for improvement.
- Convenient fitness tracking and personalized activity goals that fit her professional responsibilities and sports league commitments.
- Flexible exercise recommendations that provide alternatives to traditional gym workouts and help maintain fitness during busy periods.
- An integrated platform that encourages both financial development and physical activity without requiring excessive time or effort.

Frustrations

Daniella finds it challenging to balance her professional life, multiple sports leagues, and financial goals without over-committing her time. Although she enjoys sports, she recognizes that relying entirely on organized leagues may make it difficult to maintain a consistent fitness routine year-round. Similarly, while she has successfully developed her own investment portfolio, she wants more opportunities to test strategies and learn from her decisions without putting her savings at risk. Daniella wants a flexible and practical solution that supports her financial growth and physical well-being while fitting naturally into her lifestyle.

2.6 Priorities Assigned to Users

The project's main priority will be to focus on the beginner - novice users of the application. The goal will be to teach the basis of health/fitness and financial literacy.

2.7 User Participation

User participation will be incorporated into the development process. User feedback will help to make design decisions and provide a better scope of features.

- User interviews planned for Term 1 (the fall term custom document (CD-Fall)) to help elicit requirements and plan for the development.
- User testing on demo builds planned for Term 2 (Weeks 17 and 21) to receive feedback on decisions and provide an outside source of feedback.

2.8 Maintenance Users and Service Technicians

Standard maintenance will be the responsibility of the development team. This will include bug testing and fixing to ensure systems run as intended.

3 Mandated Constraints

3.1 Solution Constraints

The constraints below limit how the product can be built. They come from decisions recorded in the Problem Statement and the Development Plan, and the design must satisfy them.

SC-01: Android platform

Description: The product shall run as a native mobile application on Android phones.

Rationale: The team narrowed the scope from multiple operating systems to Android only so the product can be finished to a high standard within the capstone timeline. Support for iOS and other platforms is a stretch goal.

Fit Criterion: The application installs and runs all core features on an Android phone.

Related Requirements: MAS-ADPT-02

SC-02: Frontend technology

Description: The mobile client shall be written in TypeScript using the React Native framework.

Rationale: The team agreed on this stack in the Development Plan, based on the members' existing experience. React Native also leaves room for an iOS build later without rewriting the client.

Fit Criterion: All client source code is TypeScript built on React Native.

Related Requirements: COMP-LEG-07, COMP-STD-03

SC-03: Backend technology

Description: The backend application programming interface (API) shall be built in Java with the Spring framework, and persistent data shall be stored in a MySQL relational database.

Rationale: Chosen in the Development Plan. A relational database keeps user, transaction, and portfolio records structured and quick to query.

Fit Criterion: The API is a Spring application and all persistent application data lives in a MySQL database.

Related Requirements: COMP-LEG-07, COMP-STD-04, COMP-STD-05

SC-04: Fitness data sources

Description: Activity data shall be read automatically from the phone's built-in health data. Data from external devices such as fitness watches or heart rate monitors shall only be read through a software development kit (SDK) or API published by the device's manufacturer.

Rationale: Automatic tracking avoids manual entry, which users could inflate to earn currency. Using official manufacturer interfaces keeps the integration supported and within the manufacturer's terms of use.

Fit Criterion: The product has no feature for manually entering step counts, and every external device integration uses a documented SDK or API from that device's manufacturer.

Related Requirements: FUNC-03, COMP-LEG-07

SC-05: Simulated market only

Description: All trading shall take place in a simulated market using virtual currency. The product shall not handle real money, connect to a brokerage, or let users convert virtual currency into cash.

Rationale: The goal is to teach how investments grow, not to run a trading platform. Handling real money or real trades would bring financial regulation that is outside the scope of a capstone project.

Fit Criterion: No feature of the product accepts, stores, or pays out real money, and no trade is sent to a real exchange or broker.

Related Requirements: FUNC-09, FUNC-10, FUNC-17

SC-06: Distribution

Description: The product shall be distributed to users through the Android application store.

Rationale: The Problem Statement specifies that distribution is handled by the device's application store, which also handles installs and updates.

Fit Criterion: The release build meets the store's publishing requirements and can be installed from it.

Related Requirements: PROD-01, COMP-STD-01

3.2 Implementation Environment of the Current System

Flexin' is a new product and does not replace an existing system. Today, people who want to build both habits use separate tools: a fitness tracker or the phone's own health app for activity, and a stock market simulator or a real brokerage for investing. Nothing links the two.

The product will run on the environment its users already have:

- An Android phone with Health Connect, which collects activity data from the phone's sensors and from other fitness apps the user has installed.

- Optionally, a Garmin watch.
- A mobile data or Wi-Fi connection to reach the Flexin' backend.

The MySQL database (see SC-03) will be hosted on McMaster University's MySQL server, provided to the team for the course.

3.3 Partner or Collaborative Applications

The product depends on the following external applications for fitness data. Each integration must follow the provider's developer terms and permission model (see SC-04).

Health Connect (Android): The main source of step counts and workout data. The user must grant Flexin' permission to read each data type.

Garmin Connect IQ: Gives access to activity data from Garmin watches for users who own one.

The simulated market does not depend on any external price feed. Its assets and prices are fictional and are generated by the product's own market engine (see The Context of the Work). To make price movements follow real-world trends, the engine is calibrated against historical market data (see PR-PA-06). Any historical data set used for this must be free to use in an educational application.

3.4 Off-the-Shelf Software

The product will be built on the following existing software rather than writing the equivalent from scratch:

- React Native and TypeScript for the mobile client.
- Java Spring for the backend API.
- MySQL for the database.
- JUnit for backend unit and system tests, and Detox for end-to-end tests of the mobile client.

- ESLint and the linters built into the team's integrated development environments (IDEs).
- Figma for user interface (UI) mockups.
- GitHub for source control, issue tracking, and continuous integration and continuous delivery (CI/CD).

Any library added to the product must have a license that allows it to be included in proprietary software, since the project uses a proprietary license (see Section 3.8).

3.5 Anticipated Workplace Environment

Flexin' is used on the user's phone, wherever they are. That affects the design in a few ways:

- Users will often check the app in short sessions, for example after a walk or workout, while outdoors, or while moving. Key screens need to be readable on a small screen in bright light and usable with one hand.
- The connection may be slow or drop out. Activity data must not be lost when the phone is offline; it should sync once the connection returns.
- Users should not need a manual or formal training to use the app (see UHR-LRN-01). The interface has to explain financial terms itself, in line with the goal of a simple and delightful user experience.
- Step counting runs in the background through Health Connect, so the app must not drain the battery noticeably during a normal day.

3.6 Schedule Constraints

The project runs on the capstone course schedule, from September 2026 to April 2027. Dates cannot move. The milestones that constrain development are:

Week of	Milestone
Oct 19, 2026	SRS and Hazard Analysis Revision (Rev) 0; preliminary market engine working for the proof of concept
Nov 9, 2026	Virtual currency logic defined (how activity converts into currency)
Nov 23 – Dec 4, 2026	Proof of concept demo
Jan 18, 2027	Design Document Rev 0
Jan 25, 2027	Demo build ready for first round of user testing
Feb 1 – Feb 12, 2027	Rev 0 demo
Mar 1, 2027	Second round of user testing
Mar 15, 2027	Final demo
Mar 29, 2027	Final documentation (Rev 1)
Apr 5, 2027	Capstone EXPO

Features that cannot be finished by the Rev 0 demo with enough time left for user testing will be moved to the stretch goals rather than delaying the schedule.

3.7 Budget Constraints

The project has no dedicated budget. All software, services, and hosting must be free, or covered by free tiers, student plans, or resources provided by the course. This rules out paid historical market data sets and paid hosting, and any integration whose API charges per request beyond a free tier.

Development and testing will use the team members' own Android phones and the Android emulator, so no test hardware will be bought.

3.8 Enterprise Constraints

- **Course requirements.** The project must meet the deliverables and standards of the McMaster capstone course, including the document templates and checklists provided by the instructor.
- **Licensing.** The project uses a proprietary license that reserves all rights to the team (see the `LICENSE` file). This limits which third-party code can be included (see Off-the-Shelf Software).

- **Personal data.** The product stores personal and health data, such as height, weight, health conditions, and activity history. It must handle that data in line with Canadian privacy law, the Personal Information Protection and Electronic Documents Act (PIPEDA): collect only what is needed, get the user's consent, and let users see and delete their data. Details are set out in the Privacy Requirements (SEC-PRV-01 to SEC-PRV-03) and Legal Requirements (COMP-LEG-01 to COMP-LEG-04).
- **No financial advice.** The product teaches general investing concepts and must not present anything as personalized financial advice (see PR-SC-04).
- **App store policies.** The product must meet Google Play's developer policies, including its rules for health data and for finance-related apps.

4 Naming Conventions and Terminology

4.1 Glossary of All Terms, Including Acronyms, Used by Stakeholders involved in the Project

4.1.1 Terms

Term	Definition
Activity source	An app or device that records the user's physical activity: Health Connect or a Garmin watch.
Bond	A loan to a government or company that pays interest. Bonds in Flexin' are fictional.
Capstone EXPO	The public showcase of capstone projects at the end of the course.
Compounding	Growth in which returns are reinvested and earn returns of their own, so growth speeds up over time. Flexin' applies the same idea to fitness habits.
Dividend	A payment of part of a company's profits to its shareholders. Dividends in Flexin' are simulated.
Exchange-traded fund (ETF)	A fund that holds a basket of assets and is bought and sold like a single stock.
Fictional asset	A stock, bond, ETF, or fund that exists only in the Flexin' market.

Term	Definition
Fit criterion	A measurable test that shows whether a requirement has been met.
Goal progress	A user's progress towards a fitness goal, as a percentage of the goal.
Health Connect	Android's platform for sharing health and fitness data between apps.
Leaderboard group	A set of users who have opted in to be ranked against each other by goal progress.
Market engine	The part of Flexin' that generates fictional asset prices and market events.
Portfolio	The fictional assets a user holds, together with their uninvested currency.
Quiet hours	A daily period, set by the user, during which no notifications are sent.
Stock	A share of ownership in a company. Stocks in Flexin' are fictional.
Streak	The number of consecutive days or periods in which a user has met a goal.
Streak freeze	Protection a user can apply so that a missed day does not reset their streak (PR-SC-02).
Stretch goal	A feature outside the capstone scope that will be added only if time allows.
Symbolic constant	A named value defined once in Table 2 and used by name in requirements.
Virtual currency	The in-app currency, with the symbol FLX, that users earn through physical activity. It has no real-world value.
Web Content Accessibility Guidelines (WCAG) Level AA	The middle of the three conformance levels defined by WCAG.

4.1.2 Acronyms

Each acronym is spelled out where it is first used. The last column links to that place.

Acronym	Full form	Defined in
AES	Advanced Encryption Standard	Section 15.3, p. cxi
AI	Artificial intelligence	Section 1.3.1, p. viii
API	Application programming interface	Section 3.1, p. xx
BE	Business event	Section 1.3.2, p. x
BUC	Business Use Cases	Section 1.3.2, p. x
CD	Custom document	Section 2.7, p. xix
CI/CD	Continuous integration and continuous delivery	Section 3.4, p. xxiii
ETF	Exchange-traded funds	Section 1.2, p. vii
FLX	Virtual currency symbol	Section 1.3.2, p. ix
GPS	Global Positioning System	Section 6.1, p. xxx
GPT-5	Generative Pre-trained Transformer 5	Section 29.1, p. cxlix
HTTP	Hypertext Transfer Protocol	Section 15.1, p. cv
HTTPS	Hypertext Transfer Protocol Secure	Section 13.3, p. xcviii
ID	Identifier	Section 1.3.1, p. viii
IDE	Integrated development environments	Section 3.4, p. xxiii
IEEE	Institute of Electrical and Electronics Engineers	Section 17.2, p. cxxv
IP	Internet Protocol	Section 15.4, p. cxiv
ISO/IEC	International Organization for Standardization and International Electrotechnical Commission	Section 17.2, p. cxxiv
JPA	Jakarta Persistence API	Section 20.2, p. cxxix
KPI	Key performance indicators	Section 11.1, p. lxv
MIT	Massachusetts Institute of Technology	Section 20.2, p. cxxix
N/A	Not applicable	Section 1.3.2, p. ix
PDF	Portable Document Format	Section 25.1, p. cxxxvii
PHIPA	Personal Health Information Protection Act	Section 17.1, p. cxviii
PII	Personally identifiable information	Section 17.1, p. cxix
PIPEDA	Personal Information Protection and Electronic Documents Act	Section 3.8, p. xxv
PPCDA	To be confirmed by the team (see Open Issues)	—
PUC	Product Use Case	Section 8.3, p. l
Rev	Revision	Section 3.6, p. xxiv
SDK	Software development kit	Section 3.1, p. xx

Acronym	Full form	Defined in
SQL	Structured Query Language	Section 15.5, p. cxv
SR	Safety requirement	Section 1.3.2, p. ix
SRS	Software Requirements Specification	Section 1.3, p. vii
TLS	Transport Layer Security	Section 15.3, p. cxi
UI	User interface	Section 3.4, p. xxiii
UML	Unified Modeling Language	Section 1.3.4, p. xii
UX	User experience	Section 22.1, p. cxxxi
WCAG	Web Content Accessibility Guidelines	Section 4.1.1, p. xxvi
XSS	Cross-site scripting	Section 15.5, p. cxv

5 Relevant Facts And Assumptions

5.1 Relevant Facts

- Physical fitness and financial investing both rely on the principle of compounding, where small and consistent actions can produce significant long-term outcomes.
- Financial literacy among young adults remains a challenge, and many individuals lack confidence in making investment decisions without additional education.
- Existing fitness applications and investment simulators generally operate independently and do not directly connect physical activity with financial education.
- Modern Android devices are capable of collecting activity information through Health Connect and other supported fitness integrations.
- Users are increasingly familiar with gamified mobile experiences that use rewards, achievements, streaks, and leaderboards to encourage continued engagement.
- Users may possess significantly different levels of financial knowledge, fitness experience, and technological proficiency.

5.2 Business Rules

- Only validated fitness activity may be used to generate virtual currency rewards.
- Virtual currency has no real-world monetary value and cannot be exchanged for money, goods, or services. Exchanging currency for real-world rewards from sponsors is a stretch goal (see the Waiting Room).
- Virtual currency is earned only through physical activity, and a user's balance grows otherwise only through the returns on their simulated investments.
- All investment activity shall occur within a simulated market environment.
- Users must possess sufficient virtual currency before a simulated purchase can be executed.
- Users may only sell assets that they currently own within their simulated portfolio.
- Participation in social and competitive features shall be optional.
- Users retain ownership and control over their personal information and may request account deletion in accordance with applicable privacy regulations.
- Educational content provided by the application is intended for informational purposes only and shall not constitute financial advice.
- Fitness rewards shall be earned through physical activity and not through direct purchases or monetary contributions.
- Community rankings shall only include users who have elected to participate.
- User-generated content must comply with the application's community standards and may not contain offensive, abusive, discriminatory, or harassing material.

5.3 Assumptions

The following assumptions will be made during the process of development:

- **Fitness Devices Properly Track Workouts**: Flexin' will assume that data provided by fitness devices is correct and only changed through the intended means. Any artificial changes will be deemed as dishonest use and Flexin' will not provide value.
- **The User's Device Is Not Compromised**: Flexin' assumes that the device running the app has not been rooted, "jailbroken", or infected with malware.
- **The University Database Server Provides Its Advertised Guarantees**: Flexin' assumes that McMaster University's MySQL server, which hosts its database, meets its stated availability, durability, and security guarantees, so that private information is not put at risk by the server itself.
- **Users Provide Truthful Personal Information**: Flexin' assumes that the height, weight, fitness level, and health conditions users enter are accurate to the best of their knowledge.

6 The Scope of the Work

6.1 The Current Situation

The modern software landscape treats physical health management and financial education as separate domains, despite both relying on consistent daily actions and the long-term benefits of compounding. Existing fitness applications, such as Strava, Garmin Connect, and Google Fit, track physical activity through metrics such as steps, heart rate, and Global Positioning System (GPS) routes, while using badges, streaks, and milestones to encourage engagement. However, the long-term effectiveness of these incentives varies, and meaningful health benefits often require sustained participation over time **??**.

Similarly, financial simulators provide users with risk-free investment experiences but may rely on virtual capital that is disconnected from real-world

effort and financial constraints. These environments can allow users to experiment with investing without experiencing the consequences of actual financial decisions. Furthermore, their educational value depends on users' motivation to learn, rather than providing a direct incentive to develop stronger personal financial management habits.

Financial literacy remains a significant challenge in Canada. Boisclair et al. found that only 42% of Canadian respondents correctly answered three foundational questions concerning compound interest, inflation, and risk diversification **?**. The same study found that respondents who answered all three questions correctly were 10 percentage points more likely to have retirement savings, demonstrating an association between financial knowledge and retirement preparedness **?**.

Despite their shared emphasis on consistency and compounding, physical activity tracking and financial education remain largely disconnected. Fitness data is primarily managed through health-focused platforms, while investment simulators do not typically connect simulated capital to users' real-world effort or habit formation. Consequently, users lack an integrated way to explore how small, repeated actions can contribute to long-term outcomes across both domains.

This gap presents an opportunity for an integrated platform that connects physical activity with simulated investing. By linking consistent activity to incremental virtual investment opportunities, such a platform could introduce more tangible short-term incentives, demonstrate the effects of compounding, and teach financial concepts through familiar daily habits. This approach could encourage sustained participation while helping users develop the financial literacy needed for informed long-term decision-making.

6.2 The Context of the Work

The Flexin' system connects physical activity tracking with simulated investing by converting validated exercise data into virtual investment capital. Users can use this capital to build simulated portfolios containing fictional stocks, exchange-traded funds (ETFs), funds, bonds, and other supported assets without risking real money. Flexin' integrates with Android Health Connect and Garmin to retrieve supported fitness data, while its internally developed financial market engine generates fictional assets, prices, and market events. The system also provides portfolio management, financial education, and notifications.

6.2.1 Project Context Diagram

Figure 1 illustrates the interactions between Flexin' and its users, fitness integrations, and device environment. The custom financial market engine is an internal component of Flexin' that generates simulated financial data rather than retrieving real-world market prices. The engine is calibrated against historical market data so that its fictional prices follow real-world market trends (see PR-PA-06).

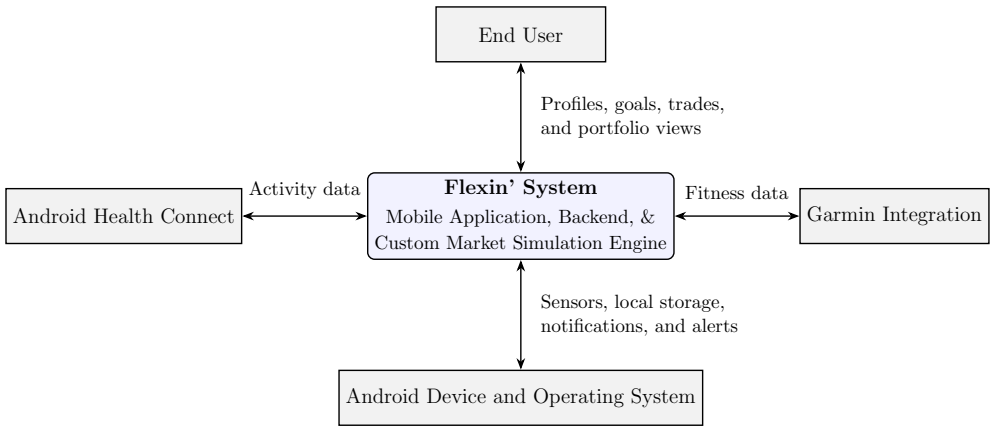


Figure 1: Flexin' Project Context Diagram

6.2.2 Adjacent Systems and External Entities

The Flexin' system interacts with the following entities and services.

Entity	Responsibility
End User	Provides profile information, exercise goals, and simulated trade instructions; receives portfolio insights, educational guidance, and virtual currency balances.
Android Health Connect	Provides aggregated health and activity data, such as step counts and exercise metrics, subject to device compatibility and user permissions.
Garmin Integration	Retrieves supported fitness and workout data from compatible Garmin devices and services, subject to API access and user authorization.
Custom Simulated Financial Market Engine	An internally developed component that creates and manages fictional stocks, ETFs, funds, bonds, and other supported assets. It generates simulated prices, market events, and asset performance without relying on real-world market data.
Android Device and Operating System	Provides the runtime environment, network connectivity, local storage, and native notification services.

Table 3: Flexin' Adjacent Systems and External Entities

6.2.3 Information Flows Across the Work Boundary

Inbound Information. Inbound information includes user commands, profile and configuration data, exercise goals, and simulated trade orders. Flexin' also receives fitness telemetry, such as step counts and workout records, through Android Health Connect and the Garmin integration. The custom financial market engine supplies internally generated asset prices, historical price series, simulated market events, and distributions such as dividends or fund returns.

Outbound Information. Outbound information includes virtual currency earned through validated activity, portfolio balances, asset allocations, and investment performance visualizations. Flexin' also provides contextual financial education on compound growth, diversification, and investment risk. Optional leaderboards, activity milestones, workout reminders, and simulated market notifications encourage continued engagement.

6.2.4 Boundary Scope and Enclosure

In Scope. Flexin' includes fitness data validation, activity-to-currency conversion, duplicate reward prevention, simulated portfolio management, and virtual trade execution. Its custom financial market engine generates fictional assets and determines their simulated prices and performance. The backend manages accounts, asset positions, transaction records, and portfolio valuations, while the React Native and TypeScript frontend presents fitness and investment dashboards. An educational rules engine provides contextual explanations of financial concepts.

Out of Scope. Flexin' does not depend on real-world stock prices, execute actual financial transactions, or manage real investment funds. It does not control the accuracy or availability of third-party fitness data, Garmin services, Android Health Connect, or device hardware. External fitness integrations remain subject to provider capabilities, API availability, device compatibility, and user permissions.

6.3 Work Partitioning

Work partitioning divides the Flexin' system into Business Use Cases (BUCs), each representing a business event that the system must handle. These use cases cover user onboarding, fitness data integration, virtual currency allocation, simulated investing, financial education, and user engagement. Table 4 summarizes the business events, their triggers, expected outcomes, and associated use cases.

Event	Business Event / Trigger	Result / Output	Associated BUC
BE-01	User registers and submits baseline goals	Account and virtual wallet initialized	BUC 1: User Onboarding
BE-02	Activity data arrives from Health Connect or Garmin	Validated activity log updated	BUC 2: Activity Ingestion
BE-03	User achieves an activity reward threshold	Virtual currency credited	BUC 3: Currency Allocation
BE-04	User submits a simulated buy order	Asset holdings and virtual balance updated	BUC 4: Asset Purchase
BE-05	User submits a simulated sell order	Holdings reduced and sale proceeds credited	BUC 5: Asset Sale
BE-06	Scheduled market simulation cycle begins	Fictional asset prices and valuations updated	BUC 6: Market Simulation
BE-07	User requests portfolio analytics	Portfolio growth and activity contributions displayed	BUC 7: Portfolio Analytics
BE-08	User requests financial guidance	Contextual educational content displayed	BUC 8: Financial Education
BE-09	User accesses community rankings	Opt-in leaderboard displayed	BUC 9: Community Leaderboard
BE-10	Reminder schedule or simulated market event is triggered	Relevant notification delivered	BUC 10: Notifications

Table 4: Flexin' Business Events and Associated Use Cases

6.4 Specifying a Business Use Case (BUC)

This section specifies each Flexin' Business Use Case (BUC), including its goal, actors, trigger, preconditions, main flow, alternative flows, and post-conditions.

BUC 1: User Onboarding

Goal: Create a user account and initialize their Flexin' profile.

Actors: End User.

Trigger: The user opens Flexin' and selects the registration option.

Preconditions: The application is available and the user does not already have an active account.

Main Flow:

1. The user submits the required registration details.
2. Flexin' validates the details and creates the account.
3. The user provides any requested baseline goals or preferences.
4. Flexin' initializes the user's profile and virtual wallet.

Alternative Flows: If required details are missing or invalid, Flexin' displays an error and requests corrections. If an account already exists, the user is prompted to sign in.

Postconditions: The user account and initial virtual wallet are created successfully.

BUC 2: Activity Ingestion

Goal: Retrieve and validate fitness activity data from supported integrations.

Actors: End User, Android Health Connect, Garmin Integration.

Trigger: New activity data becomes available or the user initiates an activity data synchronization.

Preconditions: The user is registered and has granted the required permissions to an available supported integration.

Main Flow:

1. Flexin' retrieves available activity data from Health Connect or Garmin.
2. Flexin' validates the data and checks for duplicate activities.
3. Valid activities are stored in the user's activity history.
4. The application updates the user's activity summary.

Alternative Flows: If permissions are missing, the user is prompted to grant access. If an integration is unavailable or data is invalid, Flexin' reports the issue without recording the invalid activity.

Postconditions: Valid activity data is stored without duplicate records.

BUC 3: Currency Allocation

Goal: Reward eligible physical activity with virtual currency.

Actors: End User, Flexin' Reward System.

Trigger: A validated activity satisfies a configured reward threshold.

Preconditions: The user has an active account and the activity has been validated.

Main Flow:

1. Flexin' evaluates the activity against the reward rules.
2. The system calculates the eligible virtual currency reward.
3. Flexin' credits the user's virtual wallet and records the transaction.
4. The user can view the updated balance and reward details.

Alternative Flows: If the activity does not meet the reward threshold, no currency is awarded. If the activity has already been rewarded, Flexin' prevents duplicate crediting.

Postconditions: The wallet balance and transaction ledger reflect any eligible reward.

BUC 4: Asset Purchase

Goal: Allow the user to purchase fictional financial assets using virtual currency.

Actors: End User, Flexin' Market Engine.

Trigger: The user submits a simulated buy order.

Preconditions: The user is registered, the selected asset exists, and a valid simulated price is available.

Main Flow:

1. The user selects an asset and enters the purchase quantity.
2. Flexin' calculates the simulated purchase cost.

3. The system verifies that the user has sufficient virtual currency.

4. Flexin' executes the simulated order and updates the wallet and holdings.

5. The completed transaction is recorded and displayed to the user.

Alternative Flows: If the balance is insufficient, the order is rejected. If the asset or price is unavailable, the user is informed and the transaction is not executed.

Postconditions: For a successful purchase, the virtual balance is reduced and the user's asset holdings increase accordingly.

BUC 5: Asset Sale

Goal: Allow the user to sell fictional assets and receive virtual currency.

Actors: End User, Flexin' Market Engine.

Trigger: The user submits a simulated sell order.

Preconditions: The user holds a sufficient quantity of the selected asset and a valid simulated price is available.

Main Flow:

1. The user selects an asset and enters the quantity to sell.

2. Flexin' calculates the simulated sale proceeds.

3. The system verifies that the user holds sufficient units.

4. Flexin' executes the simulated order and updates the holdings and wallet.

5. The completed transaction is recorded and displayed to the user.

Alternative Flows: If the user holds insufficient units or the asset price is unavailable, the order is rejected without changing the portfolio.

Postconditions: For a successful sale, the user's holdings decrease and the corresponding virtual currency is credited.

BUC 6: Market Simulation

Goal: Generate and update fictional asset prices and market events.

Actors: Flexin' Market Engine, Flexin' Scheduler.

Trigger: A scheduled market simulation cycle begins.

Preconditions: The market engine is operational and fictional assets have been initialized.

Main Flow:

1. The scheduler initiates a simulation cycle.
2. The engine generates price changes and fictional market events according to the system's simulation rules.
3. Flexin' updates asset valuations and relevant market information.
4. Updated prices and events become available to portfolio and trading functions.

Alternative Flows: If a simulation cycle fails, Flexin' records the failure and retains the last valid market state until an update succeeds.

Postconditions: The latest valid fictional asset prices and market events are available within Flexin'. The engine does not rely on real-world financial market data.

BUC 7: Portfolio Analytics

Goal: Present the user with portfolio performance and activity-related progress.

Actors: End User.

Trigger: The user opens the portfolio analytics view.

Preconditions: The user is registered and portfolio or activity data is available.

Main Flow:

1. Flexin' retrieves the user's virtual balance, holdings, and transaction history.
2. The system calculates portfolio value and performance metrics using simulated asset prices.
3. Flexin' retrieves relevant activity and reward information.

4. The application displays the resulting portfolio and activity summaries.

Alternative Flows: If insufficient history exists, Flexin' displays available information and indicates that more data is needed for certain metrics.

Postconditions: The user can view their latest available portfolio and activity analytics.

BUC 8: Financial Education

Goal: Provide educational guidance relevant to the user's simulated investing activity.

Actors: End User, Flexin' Educational Rules Engine.

Trigger: The user requests financial guidance or opens an educational feature.

Preconditions: The user can access the application.

Main Flow:

1. Flexin' identifies the context of the user's request.
2. The educational rules engine selects relevant educational content.
3. Flexin' presents explanations or guidance to the user.
4. The user reviews the content and may return to the application.

Alternative Flows: If no specific guidance matches the request, Flexin' presents general educational content.

Postconditions: Relevant educational content is displayed. Guidance is educational and does not execute trades on the user's behalf.

BUC 9: Community Leaderboard

Goal: Allow users who have opted in to a leaderboard group to view rankings based on progress towards their own goals or the group's agreed goals.

Actors: End User.

Trigger: The user opens the community leaderboard.

Preconditions: The user has access to Flexin' and has enabled any participation required for their own information to appear in rankings.

Main Flow:

1. Flexin' retrieves eligible ranking data from participating users.

2. The system calculates or retrieves rankings using the configured metrics.

3. The application displays the leaderboard and its ranking criteria.

Alternative Flows: If the user has opted out or no ranking data is available, Flexin' displays the available leaderboard information without including the user's opted-out data.

Postconditions: The user can view the available community rankings in accordance with participation and privacy settings.

BUC 10: Notifications

Goal: Notify users about relevant activity rewards, reminders, and simulated market events.

Actors: End User, Flexin' Notification Service, Android Device and Operating System.

Trigger: A configured reminder, eligible reward, or relevant simulated market event occurs.

Preconditions: The relevant event or schedule exists, and the user has enabled the applicable notifications and granted any required device permissions.

Main Flow:

1. Flexin' identifies the event and determines whether a notification should be sent.

2. The system generates a notification containing relevant information.

3. Flexin' sends the notification through the supported device mechanism.

4. The user receives the notification if device settings and permissions permit delivery.

Alternative Flows: If notifications are disabled or delivery fails, Flexin' does not guarantee delivery and may retain the event for in-app display.

Postconditions: The notification is submitted for delivery or the delivery failure is handled according to the system's notification rules.

7 Business Data Model and Data Dictionary

This section outlines the business data model and data dictionary for the project. The business data model provides a high-level overview of the data relevant to the business, while the data dictionary specifies the content and structure of the data elements within the system (i.e. classes, attributes, relationships). Together, these components serve as a foundation for understanding how data is organized, stored, and utilized within the business processes.

7.1 Business Data Model

This business data model is a high-level diagram of the data that is relevant to the business. It identifies the key entities and the relationships between them. The model serves as a blueprint for understanding how data flows through the business processes and how it is stored and managed. Due to still being in the earlier stages of the project, this model may be subject to changes in the later stages of the project. This business data model is not a complete representation of the data, but rather a simplified view that highlights the most important aspects of the business data. It is intended to provide a clear and concise overview of the data landscape for stakeholders and team members.

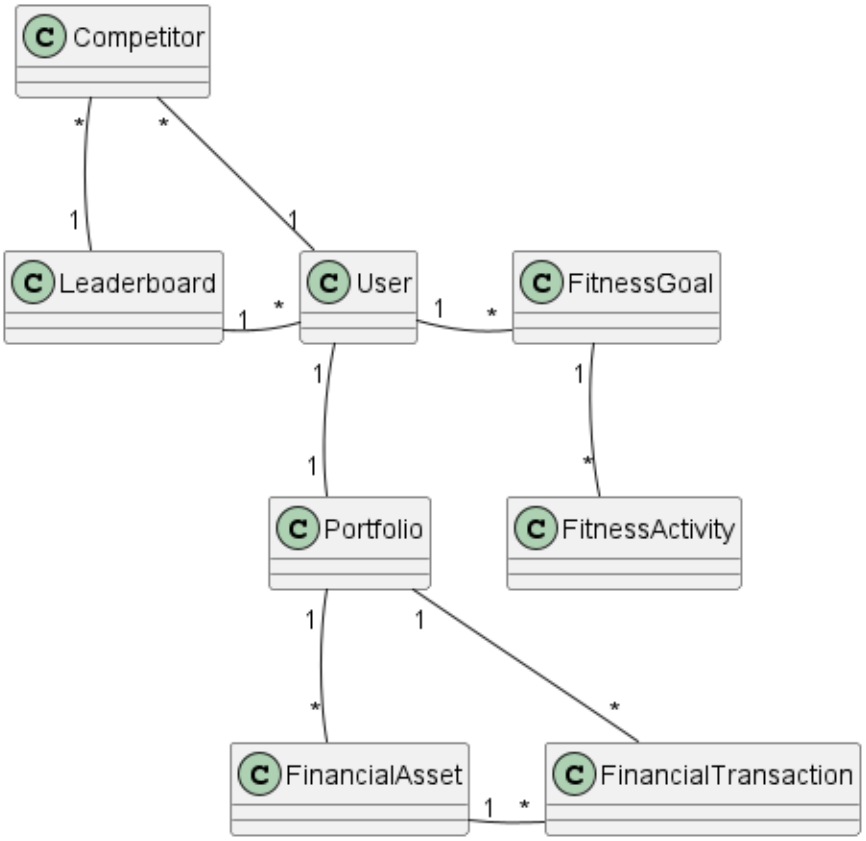


Figure 2: Business Data Model

The figure is a Unified Modeling Language (UML) class diagram. Each box is a class of business data, each line is an association between two classes, and the numbers at each end of a line are multiplicities: 1 means exactly one and * means zero or more. The attributes of each class are defined in the data dictionary.

7.2 Data Dictionary

7.2.1 User

Description: Represents a registered user of the system.

Attribute	Type	Description	Constraints

User ID	Integer	Unique identifier for a user account	Primary Key, Not Null
Email	String	Email address used to sign in and to reset the password (FUNC-01, SEC-ACC-04)	Unique, Not Null
Password	String	User authentication credential, stored as a salted hash	Hashed, Not Null
Date of Birth	Date	Used to confirm the user is 18 or older (COMP-LEG-06)	Not Null, Age $\geq$ 18
Height	Decimal	User's height, used to tailor fitness goals	Optional, Encrypted
Weight	Decimal	User's weight, used to tailor fitness goals	Optional, Encrypted
Health Conditions	String	Health conditions the user chooses to share	Optional, Encrypted
Display Alias	String	Name shown to other users on leaderboards (SEC-PRV-02)	Unique, Optional
Leaderboard Opt-In	Boolean	Whether the user has joined any leaderboard group (PR-SC-08)	Default False
Currency	Decimal	Current virtual currency balance	≥ 0
Portfolio ID	Integer	Associated portfolio identifier	Foreign Key
Experience Points	Integer	Accumulated experience points earned through platform activities	≥ 0

Active Score	Integer	Current activity score used for engagement and competition metrics	≥ 0

7.2.2 Competitor

Description: Represents a user's competitors that may appear on leaderboards and challenges.

Attribute	**Type**	**Description**	**Constraints**
Competitor ID	Integer	Unique competitor record identifier	Primary Key
User ID	Integer	Associated user account	Foreign Key
Friend Flag	Boolean	Indicates whether the competitor is on the user's friends list	True/False

7.2.3 Leaderboard

Description: Stores ranking and competitive information for users.

Attribute	**Type**	**Description**	**Constraints**
Leaderboard ID	Integer	Unique leaderboard identifier	Primary Key
User ID	Integer	Associated user on the identified leaderboard	Foreign Key
Goal Progress	Decimal	User's percentage progress towards the goal they are ranked on	≥ 0
Goal Type	Enum	Whether members are ranked on their own goals or the group's agreed goal	Own, Group

User Rank	Integer	Position within the leaderboard	≥ 1

7.2.4 Fitness Goal

Description: Represents an individual fitness objective created by a user.

Attribute	**Type**	**Description**	**Constraints**
Goal ID	Integer	Unique goal identifier	Primary Key
User ID	Integer	Identifies which user has this goal	Foreign Key
Status	Enum	Current goal state	Active, Completed, Cancelled
Reward On Completion	Decimal	Virtual currency awarded upon completion	≥ 0

7.2.5 Fitness Activity

Description: Represents a fitness activity associated with achieving a fitness goal.

Attribute	**Type**	**Description**	**Constraints**
Activity ID	Integer	Unique activity identifier	Primary Key
Goal ID	Integer	Associated fitness goal	Foreign Key
Activity Name	String	Name of activity	Not Null
Activity Details	String	What is required/associated with carrying out the activity (i.e. 10,000 steps, burn 100 calories, etc.)	Not Null

Start Time	DateTime	When the activity happened on the source device, not when it was synced (PR-PA-05)	Not Null
Source	Enum	Where the activity data came from	Health Connect, Garmin
Source Record ID	String	The source's identifier for the record, used to prevent double counting (PR-PA-03)	Unique per Source

7.2.6 Portfolio

Description: Represents a user's investment portfolio.

Attribute	Type	Description	Constraints
Portfolio ID	Integer	Unique portfolio identifier	Primary Key
Gains/Losses	Decimal	Current unrealized portfolio gain or loss	None
Asset ID	Integer	Financial asset within the identified portfolio	Foreign Key
Quantity	Integer	Quantity of asset held	≥ 0

7.2.7 Financial Asset

Description: Represents a simulated financial instrument available for trading.

Attribute	Type	Description	Constraints
Asset ID	Integer	Unique asset identifier	Primary Key
Cost To Buy/Sell	Decimal	Current purchase or sale price of the asset	≥ 0

Quantity Available	Integer	Total quantity available in the simulated market	≥ 0

7.2.8 Financial Transaction

Description: Records the purchase and sale of financial assets within a user's portfolio.

Attribute	**Type**	**Description**	**Constraints**
Transaction ID	Integer	Unique transaction identifier	Primary Key
Portfolio ID	Integer	Associated portfolio where the transaction occurred	Foreign Key
Asset ID	Integer	Asset involved in the transaction	Foreign Key
Transaction Type	Enum	Whether the transaction bought or sold the asset	Buy, Sell
Quantity	Integer	Number of units bought or sold	> 0
Transaction Cost	Decimal	Cost of the asset at the time of transaction	≥ 0
Transaction Date	DateTime	Timestamp of transaction execution	Not Null

8 The Scope of the Product

This section will explore the different features of the application. This section explores the full scope of Flexin' application and will help to define main user functions.

8.1 Product Boundary

App Functionality

The Flexin' application is designed as an educational tool to help young adults improve their personal fitness and financial literacy. The app provides a range of interactive features that allow users to engage with both domains in a meaningful and motivating way.

- Create a personal profile within the application. This profile allows users to:
    1. Enter personal information to tailor and control fitness goals.
    2. Select the desired level of difficulty and pace for financial learning modules.
- Enter, manage, and track fitness goals.
- Complete fitness challenges to earn virtual currency.
- Use virtual currency to invest in a simulated stock market environment to practice financial principles.
- Opt in to leaderboard groups that rank members by their progress towards their own goals or the group's agreed goals.

Excluded Functionality / Outside Scope

The Flexin' application will be developed specifically for the Android operating system and optimized for use within Canada. Support for other operating systems or international deployment is outside the scope of this project.

External Systems

To accurately track fitness goals, the application must retrieve data such as steps taken, heart rate, calories burned, and distance traveled. Modern Android devices provide these metrics natively, and the app will integrate with the Android Health Connect API to access this information. Additionally, many external fitness devices (e.g., Garmin watches) offer APIs and SDKs that provide detailed activity data. The Flexin' app will communicate with

these external systems to retrieve and incorporate user fitness data where supported.

8.2 Product Use Case Table

Use Case	Description
Register and setup fitness/finance goals	The user upon opening the application will be able to login to an existing account or register a new account. Upon registering a new account the user will complete a fitness and finance questionnaire to assess their current level of understanding. **See Figure 3**.
Claim Virtual Currency	Users will be able to work toward their fitness goals and complete fitness challenges (i.e. Steps/workout challenges) to collect virtual currency. Physical activity is the only way to earn new currency. **See Figure 4**.
View Portfolio and Growth Graphs	Users will have a virtual portfolio to be managed. This is where they will be able to practice their financial teachings and investment strategies. **See Figure 5**.
Use the Flexin' exchange to practice investing	"The Flexin' Exchange" will be a completely virtual stock exchange simulator that will allow users to study investment strategies and buy/sell investments. Returns on these investments are the only way, other than physical activity, for a user's balance to grow. **See Figure 6**.

Table 5: Product Use Case Table

8.3 Individual Product Use Cases (PUCs)

The flowcharts below use the following notation. A green circle marks where the use case starts, and a pink rounded box marks where it ends. Each blue

box is a screen the user sees or an action the user takes. Arrows show the order of the steps, and an italic label on an arrow is a condition or a loop, such as *Retry if needed*.

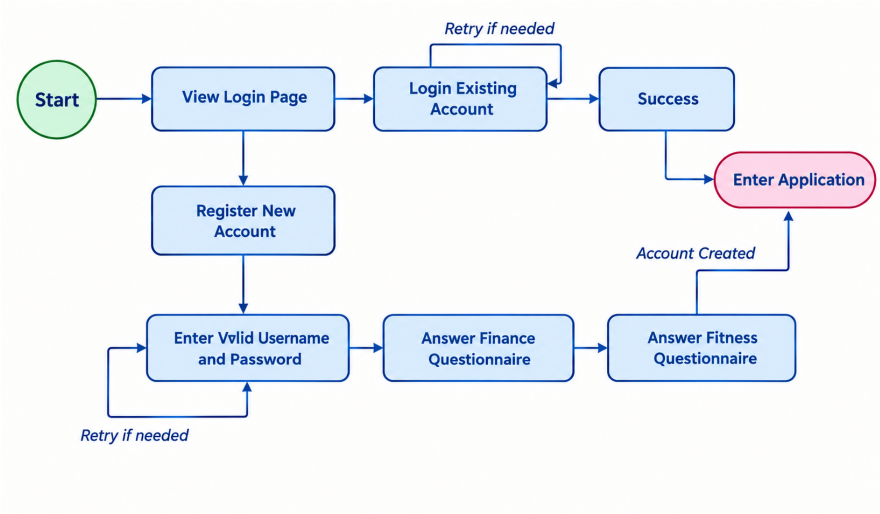


Figure 3: Product Use Case: User Onboarding

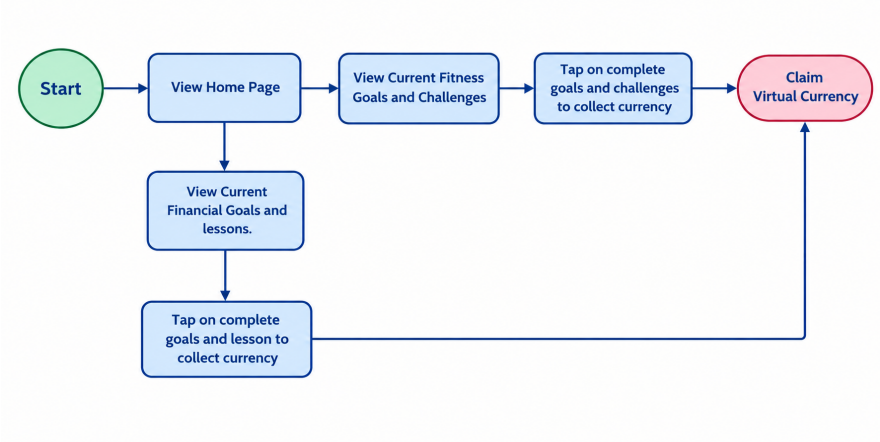


Figure 4: Product Use Case: Claim Virtual Currency

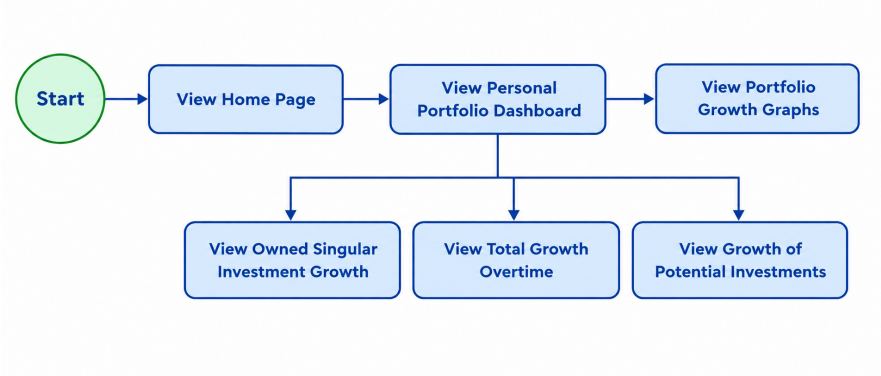


Figure 5: Product Use Case: View Stock Portfolios

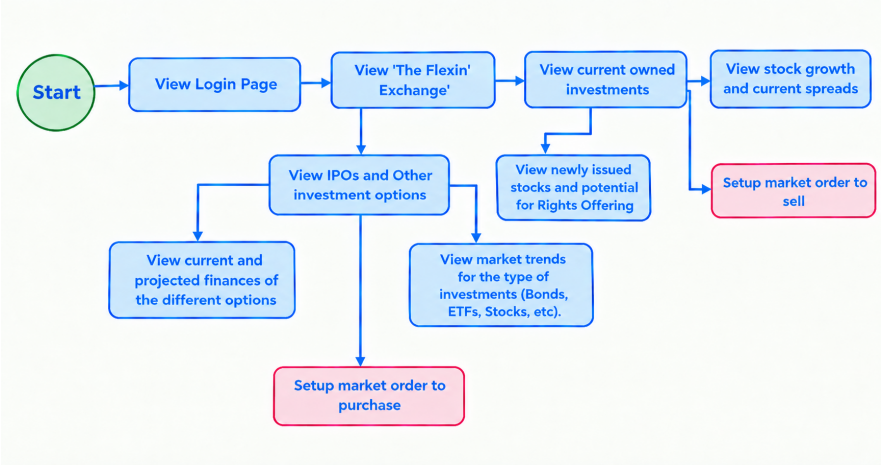


Figure 6: Product Use Case: Use the Flexin' Stock Exchange

9 Functional Requirements

This section outlines the functional requirements for the system, detailing the specific capabilities and behaviors that the system must exhibit to meet user needs and business objectives. Each requirement is identified by a unique ID, described in detail, and includes the rationale for its inclusion as well as the criteria for determining whether it has been successfully implemented.

9.1 Functional Requirements

ID	Requirement	Specification
FUNC-01	User Registration	**Description:** The system shall allow new users to create an account using a valid email address and password. **Rationale:** User accounts are required to store fitness progress, portfolio information, and competition statistics. **Fit Criterion:** A user shall be able to successfully create an account and access the application using their newly created credentials. **Related Requirements:** UHR-EOU-01, PR-SC-04, PR-SC-05, PROD-02, SEC-ACC-04, COMP-LEG-06
FUNC-02	User Authentication	**Description:** The system shall authenticate users before granting access to personalized application functionality. **Rationale:** Authentication protects personal information and ensures users can only access their own data. **Fit Criterion:** Users shall be required to provide valid credentials before accessing account-specific information. **Related Requirements:** UHR-EOU-04, SEC-ACC-01, SEC-ACC-02, SEC-ACC-04, SEC-ACC-05, SEC-IMM-02

ID	Requirement	Specification
FUNC-03	Third-Party Fitness Data Import	**Description:** The system shall import fitness activity data from supported third-party fitness providers. **Rationale:** Fitness activity data is required to reward users and track progress. **Fit Criterion:** Activities recorded through a supported third-party provider shall appear within the user's account after synchronization. **Related Requirements:** UHR-EOU-06, PR-SL-04, PR-PA-03, PR-PA-05, PR-RFT-01, PR-RFT-03, PR-SE-01, MAS-MNT-01, MAS-MNT-04, SEC-ACC-03, SEC-INT-01, COMP-LEG-01, COMP-LEG-03, DOC-02, SC-04
FUNC-04	Fitness Goal Creation	**Description:** The system shall allow users to create and manage fitness goals. **Rationale:** Goals provide motivation and measurable milestones for physical activity. This is also how users will earn virtual currency to invest in the simulated stock market. **Fit Criterion:** Users shall be able to create, modify, and delete fitness goals. **Related Requirements:** UHR-PER-01, UHR-LRN-03, UHR-ACC-06, PR-SC-06

ID	Requirement	Specification
FUNC-05	Fitness Goal Tracking	**Description:** The system shall track user progress toward active fitness goals. **Rationale:** Progress tracking enables users to monitor their fitness achievements. This also allows the system to determine when a user has completed a goal and is eligible for virtual currency rewards. **Fit Criterion:** Goal progress shall automatically update when related activities are synchronized. **Related Requirements:** UHR-INT-01, PR-SC-02, PR-PA-05
FUNC-06	Reward Allocation	**Description:** The system shall award virtual currency when users complete eligible fitness activities or goals. **Rationale:** Virtual rewards act as the primary incentive linking physical activity to financial education. **Fit Criterion:** Virtual currency balances shall increase following successful completion of reward-eligible activities. **Related Requirements:** UHR-UND-03, PR-SC-01, PR-PA-02, PR-PA-03, SEC-INT-01, SEC-AUD-01, SEC-IMM-03
FUNC-07	Virtual Currency Tracking	**Description:** The system shall maintain a record of each user's current virtual currency balance. **Rationale:** Users require visibility into available funds for portfolio management activities. **Fit Criterion:** Users shall be able to view their current virtual currency balance at any time. **Related Requirements:** LF-STY-02, PR-PA-01, PR-RFT-04

ID	Requirement	Specification
FUNC-08	Portfolio Creation	**Description:** The system shall create and maintain a portfolio for each registered user. **Rationale:** Portfolios provide users with a mechanism for applying virtual currency toward investment activities. **Fit Criterion:** A portfolio shall automatically be created when a user account is established. **Related Requirements:** PR-RFT-02
FUNC-09	Asset Purchase	**Description:** The system shall allow users to purchase supported financial assets using virtual currency. **Rationale:** Asset purchases are necessary to simulate investment decision-making and portfolio growth. **Fit Criterion:** Users shall be able to purchase available assets provided they possess sufficient virtual currency. **Related Requirements:** UHR-LRN-02, UHR-UND-03, PR-SL-03, PR-SC-03, PR-RFT-02, PR-SE-02, SEC-INT-02, SEC-IMM-02, SC-05
FUNC-10	Asset Sale	**Description:** The system shall allow users to sell previously purchased financial assets and realize gains or losses. **Rationale:** Users must be able to modify portfolio holdings and realize gains or losses. **Fit Criterion:** Users shall be able to sell assets currently held within their portfolio. **Related Requirements:** UHR-UND-03, PR-SL-03, PR-RFT-02, SEC-INT-02, SC-05

ID	Requirement	Specification
FUNC-11	Transaction Recording	**Description:** The system shall record all financial transactions executed by users. **Rationale:** Transaction records provide accountability and support portfolio calculations. **Fit Criterion:** Every asset purchase and sale shall generate a corresponding transaction record that can be viewed by the user at any time. **Related Requirements:** PR-C-02, SEC-AUD-01
FUNC-12	Portfolio Performance Calculation	**Description:** The system shall calculate portfolio gains and losses based on current asset values. **Rationale:** Portfolio performance provides feedback regarding investment decisions and how the user can adjust their strategies moving forward. **Fit Criterion:** Users shall be able to view current portfolio value and unrealized gains or losses. **Related Requirements:** LF-APP-02, UHR-UND-02, PR-PA-04, PR-PA-06, PR-RFT-04

ID	Requirement	Specification
FUNC-13	Leaderboard Participation	**Description:** The system shall let users join and leave leaderboard groups, and shall show each group's members ranked by their progress towards their own goals or towards the goals the group has agreed on. Ranking rules are constrained by PR-SC-08. **Rationale:** Competitive features encourage continued engagement and motivation. Ranking by progress towards personal or agreed goals keeps the competition fair between users of different fitness and financial levels. **Fit Criterion:** A user can join a group, view its ranking, and leave it. Each member's position is computed from their percentage progress towards their own or the group's agreed goal and updates when their progress changes. **Related Requirements:** UHR-PER-03, UHR-PER-04, UHR-POL-04, PR-SC-08, SEC-PRV-02, SEC-IMM-03
FUNC-14	Friend Management	**Description:** The system shall allow users to create and manage lists of friends they wish to compete with. **Rationale:** Users may wish to compare progress with selected friends rather than with unknown users. **Fit Criterion:** Users shall be able to add and remove friends from their friends list. **Related Requirements:** UHR-PER-04, UHR-POL-05

ID	Requirement	Specification
FUNC-15	Private Leaderboards	**Description:** The system shall allow users to create private leaderboard groups so they can invite friends to compete. Private groups follow the same leaderboard rules as public ones (PR-SC-08), so friends cannot use them to earn currency. **Rationale:** Users may wish to compare progress with selected friends rather than with unknown users. **Fit Criterion:** Users shall be able to create private competition groups with select friends by inviting them. **Related Requirements:** UHR-PER-04, PR-SC-08
FUNC-16	Progress Reporting	**Description:** The system shall provide users with summaries of fitness activity, goal completion, virtual currency earnings, portfolio performance, and leaderboard standings. **Rationale:** Summaries help users understand their overall progress and achievements. **Fit Criterion:** Users shall be able to access current and historical progress information at any time. **Related Requirements:** LF-APP-01, UHR-EOU-02, PR-RFT-04, PR-C-02, PR-L-02

10 Look and Feel Requirements

10.1 Appearance Requirements

ID	Requirement	Specification
LF-APP-01	Visual Hierarchy	**Description:** The dashboard shall present fitness metrics and financial metrics in a balanced, uncluttered layout using unified summary cards. **Rationale:** Equal emphasis reinforces the relationship between physical activity and simulated investment growth. **Fit Criterion:** In user testing, at least 80% of participants identify both their current fitness progress and their portfolio value from the dashboard within 5 seconds. **Related Requirements:** FUNC-16, UHR-EOU-02
LF-APP-02	Compounding Chart	**Description:** The portfolio chart shall distinguish cumulative workout earnings from simulated investment gains and dividends. **Rationale:** Users should understand the difference between contributions and compounded returns. **Fit Criterion:** Two visually distinct chart traces shall allow users to distinguish these values within 5 seconds. **Related Requirements:** FUNC-12, UHR-ACC-03
LF-APP-03	Responsive Layout	**Description:** The interface shall adapt to Android screen sizes, aspect ratios, pixel densities, and system font scaling. **Rationale:** Flexin' must remain usable across supported Android devices. **Fit Criterion:** On Android 10+ devices with 5.5–6.8-inch screens, no text shall clip or touch targets overlap, including at 130% font scaling. **Related Requirements:** UHR-ACC-02, UHR-ACC-04, PR-L-01, PROD-02

ID	Requirement	Specification
LF-APP-04	Theme and Contrast	**Description:** The interface shall support Light and Dark modes with distinguishable fitness and financial status indicators. **Rationale:** Consistent contrast improves readability and accessibility in different environments. **Fit Criterion:** Both Light and Dark modes meet the colour contrast ratios required by COMP-STD-02, and fitness and financial status indicators remain distinguishable in both modes. **Related Requirements:** UHR-ACC-03, COMP-STD-02
LF-APP-05	Navigation	**Description:** A persistent bottom navigation bar shall link to Dashboard, Activity, Market/Portfolio, Learn, and Community. **Rationale:** Simple navigation reduces user effort. **Fit Criterion:** The bottom navigation bar with these five destinations is visible on every primary screen. How many taps primary operations may take is specified in UHR-EOU-03. **Related Requirements:** UHR-EOU-03

10.2 Style Requirements

ID	Requirement	Specification
LF-STY-01	Approachable Tone	**Description:** Typography, interface styling, and copy shall be welcoming and accessible. Financial terminology shall use clear language with contextual tooltips. **Rationale:** Reducing intimidation encourages financial learning and exploration. **Fit Criterion:** At least 80% of target-demographic evaluators shall rate the interface as approachable, supportive, or inviting. **Related Requirements:** UHR-UND-01, UHR-POL-01
LF-STY-02	Virtual Currency Symbol	**Description:** All amounts of virtual currency shall be shown with the virtual currency symbol (FLX), never with a real-world currency symbol. **Rationale:** Users must not confuse Flexin' with a real brokerage or financial account. **Fit Criterion:** No screen shows a virtual amount with a real-world currency symbol such as $, and every virtual amount uses FLX. The label marking the market as simulated is specified in PR-SC-03. **Related Requirements:** FUNC-07, PR-SC-03

ID	Requirement	Specification
LF-STY-03	Non-Intrusive Gamification	**Description:** Milestone rewards shall use subtle, encouraging animations without disrupting normal use. **Rationale:** Rewards should support sustainable habits rather than distract users. **Fit Criterion:** Reward animations shall last no longer than 1.5 seconds and be immediately dismissible by tapping the background or swiping. **Related Requirements:** UHR-POL-02, PR-SL-02
LF-STY-04	Component Consistency	**Description:** Buttons, forms, modals, tooltips, cards, and toggles shall follow a shared styling system. **Rationale:** Consistency builds trust and reduces the learning curve. **Fit Criterion:** A design review of every screen finds no button, form, modal, tooltip, card, or toggle that departs from the shared style guide. **Related Requirements:** UHR-EOU-07

11 Usability and Humanity Requirements

This section outlines the usability and humanity requirements for the system, including ease of use, personalization and internationalization, learning, understandability and politeness, and accessibility.

11.1 Ease of Use Requirements

ID	Requirement	Specification
UHR-EOU-01	Onboarding Completion Time	**Description:** The system shall enable new users to complete initial account setup and onboarding within 5 minutes. **Rationale:** Users should be able to begin using the application almost immediately without excessive setup requirements. Within 5 minutes allows users to maintain their engagement with the system. If the setup and onboarding take too long, users may become frustrated and disengaged. **Fit Criterion:** A first-time user shall be able to setup their account, complete onboarding and reach the main dashboard/screen within 5 minutes. This will be tested later in the project through user testing and observation. **Related Requirements:** FUNC-01, UHR-LRN-01

ID	Requirement	Specification
UHR-EOU-02	Dashboard Accessibility	**Description:** The system shall provide direct access to fitness progress, portfolio information, and rewards through the main dashboard. **Rationale:** Users should not need to navigate through multiple screens to access frequently used information. **Fit Criterion:** Important high-level key performance indicators (KPIs) pertaining to fitness progress, portfolio value, and virtual currency balance shall be visible from the application's primary screen after login. Users should be able to answer questions like "How much currency do I have available?", "What percentage of my fitness goal is complete?" and "How is my portfolio performing?" within 1 minute after logging into the system. **Related Requirements:** FUNC-16, LF-APP-01
UHR-EOU-03	Minimal Screen Navigation	**Description:** The system shall minimize navigation complexity by ensuring that portfolio management and lower-level metric viewing can be reached from the main dashboard within two taps. **Rationale:** Users should not need to navigate through multiple screens to access detailed metrics and adjust their portfolios. Excessive navigation increases user frustration and decreases engagement. **Fit Criterion:** Users shall be able to access detailed fitness statistics, portfolio management, virtual currency transactions, and goal tracking within two taps from the main dashboard. **Related Requirements:** LF-APP-05

ID	Requirement	Specification
UHR-EOU-04	Error Feedback and Recovery	**Description:** The system shall provide clear, actionable, and immediate feedback to the user whenever a user action cannot be completed, including failed account creation and sign-in. **Rationale:** Clear feedback helps users understand how to resolve issues and continue using the application, instead of being locked out or stuck with no explanation. **Fit Criterion:** Every error shown during testing appears immediately, states what went wrong, and gives at least one action the user can take to fix it. **Hazard Analysis:** SR4 (H1.1) **Related Requirements:** FUNC-02, MAS-SUP-01
UHR-EOU-05	Positive Feedback	**Description:** The system shall provide immediate visual feedback following user actions to confirm whether the requested operation was completed successfully. **Rationale:** Users should be able to easily determine whether an action has been executed correctly without needing to manually verify the system state. Clear feedback increases user confidence and reduces confusion. **Fit Criterion:** All user-initiated actions, including goal updates, portfolio transactions, account modifications, and fitness device connections, shall display a confirmation message, notification, or visual state change within 2 seconds of successful completion. **Related Requirements:** UHR-UND-03, PR-SL-02, PR-SL-03

ID	Requirement	Specification
UHR-EOU-06	Simple and Reliable Device Connectivity	**Description:** The system shall simplify the process of connecting supported fitness devices and applications and maintain strong and reliable connections to these devices. **Rationale:** Integration with existing fitness technologies is a core project goal. Users should be able to link their existing devices without requiring technical expertise. Once connected, the system shall maintain a stable connection to ensure continuous data synchronization so that the user is not constantly required to re-establish the connection - as this will cause user frustration. **Fit Criterion:** Users shall be able to perform a one-time action to connect a supported fitness data source through a guided setup workflow. Once connected the first time, the system shall not sever this connection more than three times per year without explicit user intervention. **Related Requirements:** FUNC-03, DOC-02
UHR-EOU-07	Consistent User Interface	**Description:** The system shall maintain consistent visual design, terminology, and navigation patterns across all screens. **Rationale:** Consistency improves user familiarity and reduces the effort required to learn new functionality. **Fit Criterion:** A design review checklist covering icons, terminology, colours, and interaction patterns finds no inconsistency between screens for the same action. **Related Requirements:** LF-STY-04

ID	Requirement	Specification
UHR-EOU-08	Learnability	**Description:** The system shall require no more than 10 minutes for users to become proficient in performing core tasks and navigating the application. **Rationale:** Users should not require an excessive amount of time to learn and use the application effectively. Excessive time required to learn how to use the core functionalities of the application may lead to user frustration and disengagement. **Fit Criterion:** First-time users shall be able to make their first investment, set one fitness goal, and view fitness and portfolio performance metrics within 10 minutes from landing on the main screen after login. **Related Requirements:** UHR-EOU-09, UHR-LRN-01, TRN-01, TRN-02

ID	Requirement	Specification
UHR-EOU-09	Efficient Re-Learnability	**Description:** The system shall require no more than 5 minutes for former users to re-learn how to execute core tasks and navigating the application. **Rationale:** Even after time away from the system, users should be able to pick back up right where they left off. If a user has to spend too much time re-learning the application, it may lead to frustration and discourage users from re-engaging with the system. **Fit Criterion:** Former users with at least 1 month of consistent daily usage shall be able to re-learn how to manage at least one investment, manage at least one fitness goal, and view fitness and portfolio performance metrics within 5 minutes of returning to the system after 6+ months of inactivity. **Related Requirements:** UHR-EOU-08

11.2 Personalization and Internationalization Requirements

ID	Requirement	Specification
UHR-PER-01	Personalized Fitness Goals	**Description:** The system shall allow users to configure fitness goals based on their personal fitness level and preferences. **Rationale:** Users will possess different fitness capabilities and activity levels. Personalized goals encourage engagement while remaining achievable. **Fit Criterion:** Users shall be able to modify daily and weekly fitness goals that will take effect the next time period. **Related Requirements:** FUNC-04
UHR-PER-02	Customizable Notifications	**Description:** The system shall allow users to configure which notifications they receive and how frequently they are delivered. **Rationale:** Users have varying preferences regarding reminders and activity notifications. **Fit Criterion:** Users shall be able to independently enable or disable workout reminders, portfolio updates, and achievement notifications. **Related Requirements:** UHR-ACC-05, PR-SC-07
UHR-PER-03	Customizable Privacy Settings	**Description:** The system shall allow users to control the visibility of their fitness achievements, portfolio statistics, and leaderboard participation. **Rationale:** Users have different privacy expectations and comfort levels regarding public sharing of personal information. **Fit Criterion:** Users shall be able to independently configure visibility settings for all community-facing features that will take immediate effect. **Related Requirements:** FUNC-13

ID	Requirement	Specification
UHR-PER-04	Customizable Competition Groups	**Description:** The system shall allow users to customize the set of users with whom they share and compare fitness and portfolio progress. **Rationale:** Users may have different comfort levels regarding competition and sharing personal achievements. Allowing users to select their competition group creates a more personalized and engaging experience where friends can motivate each other. **Fit Criterion:** Users shall be able to choose between not participating (the default), joining a private friends-only group, and joining the public community-wide leaderboard. **Related Requirements:** FUNC-13, FUNC-14, FUNC-15
UHR-INT-01	Canadian Time Zone Support	**Description:** The system shall automatically detect and utilize the user's local Canadian time zone when calculating daily goals, rewards, and streaks. **Rationale:** Users may access the application from any Canadian province or territory. **Fit Criterion:** Daily statistics and streak calculations shall correspond to the user's local device time zone. **Related Requirements:** FUNC-05, PR-PA-05, CUL-01

11.3 Learning Requirements

ID	Requirement	Specification
UHR-LRN-01	Minimal User Training	**Description:** The system shall enable first-time users to successfully use the application's core functionality without requiring formal training. **Rationale:** The application targets users with varying expertise and should be intuitive for a broad audience. **Fit Criterion:** At least 80% of first-time users shall be able to synchronize fitness data, earn virtual currency, and purchase an investment asset without external assistance. **Related Requirements:** UHR-EOU-01, UHR-EOU-08, TRN-01
UHR-LRN-02	Learning Curve of Financial Concepts	**Description:** The system shall present investment concepts in a manner that can be understood by users with little or no prior experience. **Rationale:** One of the system's primary goals is to educate users on financial investing. This means that all users, regardless of their prior financial experience, should be capable of learning and understanding the financial concepts presented. **Fit Criterion:** During user testing, at least 80% of first-time investors shall successfully complete their first virtual investment and view their portfolio's growth after one week without assistance. **Related Requirements:** FUNC-09, UHR-LRN-04, DOC-01

ID	Requirement	Specification
UHR-LRN-03	Learning Curve of Fitness Based Concepts	**Description:** The system shall present fitness concepts in a manner that can be understood by users with little or no prior experience. **Rationale:** One of the system's primary goals is to educate users on fitness and health related activities. This means that all users, regardless of their prior fitness experience, should be capable of learning and understanding the fitness concepts presented. **Fit Criterion:** During user testing, at least 80% of first-time participants in a new fitness activity shall successfully accomplish their set goal and view their health metrics after one week without assistance. **Related Requirements:** FUNC-04, UHR-LRN-04

ID	Requirement	Specification
UHR-LRN-04	Integrated Educational Assistance	**Description:** The system shall provide a plain-language explanation, reachable directly from the screen where it appears, for every financial term, investment metric, fitness metric, and health term shown to users. Explanations shall not assume any prior investing or fitness knowledge. **Rationale:** Users have very different levels of financial and fitness knowledge, and many target users have little experience with investing. Explaining each term where it appears lets them learn while they use the app instead of leaving to look things up. **Fit Criterion:** A review of every screen finds an explanation for each financial and fitness term or metric shown. In user testing, at least 80% of participants rate the explanations as clear (4 or 5 on a 5-point scale). **Related Requirements:** UHR-LRN-02, UHR-LRN-03, UHR-UND-01, CUL-02, COMP-LEG-05, DOC-01

11.4 Understandability and Politeness Requirements

ID	Requirement	Specification
UHR-UND-01	Plain Language Communication	**Description:** The system shall present information using clear and concise language appropriate for users with limited financial and fitness knowledge. **Rationale:** The system targets users with varying levels of experience in personal finance and physical fitness. Clear language improves comprehension and engagement. **Fit Criterion:** User-facing content shall avoid unexplained technical terminology and provide explanations for all specialized terms. **Related Requirements:** LF-STY-01, UHR-LRN-04, CUL-01, CUL-02
UHR-UND-02	Compounding Transparency and Distinction	**Description:** The system shall clearly distinguish between currency gains through the completion of fitness-related activities and gains resulting from portfolio performance. **Rationale:** One of the primary educational goals of the application is to demonstrate the concept of compounding over time. Users should not focus solely on gaining currency through fitness-related activities - nor should they solely focus on portfolio performance. The system should clearly communicate the distinction between these two sources of currency gains to ensure users understand the impact of both fitness and investment decisions on their overall success. **Fit Criterion:** Portfolio views shall separately display user-earned virtual currency contributions and investment gains or losses. **Related Requirements:** FUNC-12

ID	Requirement	Specification
UHR-UND-03	Action Outcome Transparency	**Description:** The system shall clearly communicate the outcome of actions affecting fitness progress, virtual currency balances, and portfolio holdings. **Rationale:** Users should understand how their actions impact system state and long-term progress at all times. **Fit Criterion:** Following any successful transaction or activity synchronization, the system shall immediately identify the resulting changes to affected data. **Related Requirements:** FUNC-06, FUNC-09, FUNC-10, UHR-EOU-05
UHR-POL-01	Respectful User Communication	**Description:** The system shall communicate with users in a respectful and professional manner at all times. **Rationale:** Positive interactions improve user satisfaction and promote long-term engagement with the application. **Fit Criterion:** User-facing notifications, prompts, and messages shall **never** contain insulting, belittling, or derogatory language. **Related Requirements:** LF-STY-01, UHR-POL-03, UHR-POL-04, CUL-02
UHR-POL-02	Positive Motivational Feedback	**Description:** The system shall provide encouraging feedback regarding fitness activity and portfolio performance. **Rationale:** Long-term engagement is a primary project goal and positive reinforcement supports continued participation. **Fit Criterion:** Achievement notifications and milestone messages shall always use positive and supportive language. **Related Requirements:** LF-STY-03, PR-SC-07

ID	Requirement	Specification
UHR-POL-03	Non-Punitive Performance Messaging	**Description:** The system shall avoid language that shames users for missed goals, inactivity, or poor portfolio performance. **Rationale:** Users should remain motivated even when progress is temporarily reduced. Users may experience setbacks but should not be discouraged from continuing. **Fit Criterion:** Notifications regarding missed goals or declining portfolio performance shall focus on future opportunities for improvement rather than negative judgement. During user testing, a survey will be sent out to users to gauge their satisfaction with the feedback provided when they encounter performance challenges - it is expected that survey results will return that no more than 10% of users were discouraged to continue after failing to meet a fitness and/or financial goal. **Related Requirements:** UHR-POL-01, PR-SC-07
UHR-POL-04	Respectful Competitive Features	**Description:** The system shall present leaderboard rankings and competitive information in a respectful and encouraging manner. **Rationale:** Users may be motivated by competition, but competitive features should not create a negative user experience. **Fit Criterion:** Leaderboard interfaces shall not contain derogatory labels or language that demeans lower-ranked users. **Related Requirements:** FUNC-13, UHR-POL-01

ID	Requirement	Specification
UHR-POL-05	Respectful User-Generated Content	**Description:** The system shall prevent users from creating usernames, profile information, or public content containing offensive, discriminatory, harassing, or abusive language. **Rationale:** The system should foster a welcoming and respectful environment for all users participating in community and competitive features. No users should be subject to offensive or derogatory content when competing with other users online. **Fit Criterion:** Attempts to create or modify public-facing content containing prohibited language shall be rejected and the user shall be notified that the content violates community guidelines. **Related Requirements:** FUNC-14

11.5 Accessibility Requirements

ID	Requirement	Specification
UHR-ACC-01	Screen Reader Compatibility	**Description:** The system shall support screen reader technologies for all primary application workflows. **Rationale:** Users with visual impairments may rely on screen readers to navigate the application. Users with visual impairments should not be excluded from benefitting from this application. **Fit Criterion:** All interactive interface elements shall have descriptive accessibility labels that can be read aloud. **Related Requirements:** COMP-STD-02

ID	Requirement	Specification
UHR-ACC-02	Scalable Text Support	**Description:** The system shall support device-configured text scaling without loss of functionality. **Rationale:** Users with reduced vision may require larger text sizes to comfortably use the application. **Fit Criterion:** All application screens shall remain usable when the device text size is increased to the maximum supported accessibility setting. **Related Requirements:** LF-APP-03, COMP-STD-02
UHR-ACC-03	Color Independent Information	**Description:** The system shall not rely solely on color to communicate information or system status. **Rationale:** Users with color vision deficiencies may be unable to distinguish information represented exclusively through color. **Fit Criterion:** All status indicators, icons, buttons, graphs, and notifications shall include a secondary visual indicator in addition to color. **Related Requirements:** LF-APP-02, LF-APP-04, COMP-STD-02
UHR-ACC-04	Touch Target Accessibility	**Description:** The system shall provide interface controls that are sufficiently large for reliable touch interaction. **Rationale:** Users with limited dexterity or motor impairments may have difficulty selecting small interface elements. **Fit Criterion:** All touch-interactive elements shall meet the minimum touch target size recommended by Android accessibility guidelines. **Related Requirements:** LF-APP-03, COMP-STD-02

ID	Requirement	Specification
UHR-ACC-05	Alternative Notification Methods	**Description:** The system shall signal important events (a completed goal, currency earned, a failed sync, and security alerts) in at least two ways: visually, and by sound or vibration. **Rationale:** Users who are deaf or hard of hearing may not perceive audio notifications. Users with visual impairments may not be able to perceive visual notifications. Providing alternative notification methods ensures that all users can stay informed of important events. **Fit Criterion:** In testing, each important event is signalled visually and by sound or vibration, and with the device's sound turned off the visual and vibration signals still occur. **Related Requirements:** UHR-PER-02, COMP-STD-02
UHR-ACC-06	Inclusive Activity Tracking	**Description:** The system shall not require a single type of physical activity as the sole method of earning rewards. **Rationale:** Users with disabilities, injuries, temporary medical conditions, or differing fitness abilities may be unable to participate in certain forms of exercise. **Fit Criterion:** The system shall support multiple reward-eligible activity types and allow users to select activities appropriate to their abilities. **Related Requirements:** FUNC-04, MAS-MNT-01

12 Performance Requirements

Unless stated otherwise, the timing requirements in this section are measured on a mid-range Android phone (one released within the last four years) with a typical mobile data or Wi-Fi connection.

12.1 Speed and Latency Requirements

ID	Requirement	Specification
PR-SL-01	App launch	**Description:** The app shall show the home screen within 3 seconds of being opened from a closed state. **Rationale:** Users will mostly open the app for short check-ins, such as after a walk or workout. A slow start makes those check-ins less likely. **Fit Criterion:** Across 20 cold starts on the reference phone, the home screen appears in 3 seconds or less in at least 19 of them. **Related Requirements:** PR-C-01
PR-SL-02	Response to input	**Description:** The app shall respond to a tap or navigation action within 1 second. **Rationale:** Supports the goal of a simple and delightful user experience. Delays longer than about a second make an app feel broken. **Fit Criterion:** At least 95% of measured screen transitions and button responses complete within 1 second. **Related Requirements:** LF-STY-03, UHR-EOU-05, PR-C-01, PR-C-02, PR-SE-03

ID	Requirement	Specification
PR-SL-03	Trades	**Description:** After a user confirms a buy or sell order, the updated holdings and currency balance shall be shown within 2 seconds. **Rationale:** Users need to see the result of a trade straight away to connect the action with its effect on their portfolio. **Fit Criterion:** At least 95% of test trades show the updated portfolio within 2 seconds of confirmation. **Related Requirements:** FUNC-09, FUNC-10, UHR-EOU-05, PR-C-01, PR-SE-03
PR-SL-04	Activity sync	**Description:** When the app is opened, activity recorded since the last sync shall be read and turned into currency within 10 seconds. **Rationale:** The reward has to feel tied to the activity that earned it. The whole point of the product is to make a distant payoff feel immediate. **Fit Criterion:** In testing, the currency balance reflects all activity available in Health Connect within 10 seconds of opening the app. **Related Requirements:** FUNC-03

12.2 Safety-Critical Requirements

Flexin' does not control physical equipment, so none of its failures can directly injure someone. The Hazard Analysis does identify ways the product could encourage harm, and the requirements below address them. Each requirement lists the safety requirements and hazards from the Hazard Analysis that it traces to.

ID	Requirement	Specification
PR-SC-01	Daily earning limits	**Description:** The currency a user can earn from activity in one day shall not exceed DailyEarnCap, and activity beyond HealthyDailyActivity in a day shall earn currency at a reduced rate. **Rationale:** Without these limits, the product would reward users for exercising past safe limits, or while injured, to earn more. **Fit Criterion:** No amount of activity recorded in a single day earns more than DailyEarnCap, and activity beyond HealthyDailyActivity earns less per unit than activity below it. **Hazard Analysis:** SR8 (H4.1, H5.1) **Related Requirements:** FUNC-06
PR-SC-02	No penalties for missed days	**Description:** The product shall not take away currency or holdings when a user misses a goal or a day of activity. Users shall be able to freeze their streak for days missed because of illness, injury, rest, or a delayed sync. **Rationale:** The Hazard Analysis lists shame over missed goals as a psychological hazard. Losing earned progress or a long streak would make that worse, could push users to exercise while injured or unwell, and makes users feel they have failed through no fault of their own. **Fit Criterion:** No feature of the product reduces a user's balance or holdings because of inactivity, and a user can freeze their streak so that a missed day does not reset it. **Hazard Analysis:** SR8 (H3.1, H4.1, H11.1) **Related Requirements:** FUNC-05

ID	Requirement	Specification
PR-SC-03	Simulated market is labelled	**Description:** Every screen that shows prices, holdings, or returns shall state that the market is simulated and that results do not predict real returns. Each of these screens shall link to an explanation of what the simulation does not model, such as fees, taxes, and real-world volatility. **Rationale:** The Hazard Analysis lists financial loss from users applying strategies from the app to real markets. Users who treat simulated results as real-world truth may lose real money. Currency amounts on these screens use the FLX symbol (LF-STY-02). **Fit Criterion:** A review of every market and portfolio screen finds the notice visible on each one, and the explanation of the simulation's limits can be reached from each one. **Hazard Analysis:** SR11 (H6.1, H7.1) **Related Requirements:** FUNC-09, LF-STY-02

ID	Requirement	Specification
PR-SC-04	Not financial advice	**Description:** The product shall state that its content is for education only and is not financial advice. The statement shall be shown during sign-up, where the user must acknowledge it before continuing, and shall stay available from the settings screen. Lessons, tips, portfolio insights, and any other content that discusses investing shall not recommend specific real-world investments to a user. **Rationale:** The Hazard Analysis lists financial loss from users acting on what they learn in the app. Flexin' is not a licensed financial advisor, so users must understand that nothing in the app is personal advice about their real money. The statement also protects the team from legal claims over losses users make in real markets, so it is a legal requirement as well as a safety one. **Fit Criterion:** A new account cannot be created without acknowledging the statement, the statement can be reached from settings, and a review of all educational content finds no recommendations to buy or sell specific real-world investments. **Hazard Analysis:** SR11 (H6.1, H7.1) **Related Requirements:** FUNC-01

ID	Requirement	Specification
PR-SC-05	Exercise injury warning	**Description:** The product shall warn users that physical activity carries a risk of injury, that they should exercise within their own limits, and that they should consult a doctor before starting a new exercise routine if they have health concerns. The product's terms of use shall state that users exercise at their own risk and that the Flexin' team is not responsible for any injury that happens while using the app. The user must acknowledge both during sign-up before continuing. **Rationale:** The product rewards physical activity, so some users will exercise more because of it. The Hazard Analysis assumes users only attempt exercises they are able to do safely, and places injuries from exercise outside the product's control. Users need to be told this directly. The responsibility statement is a legal requirement as well as a safety one, and its exact wording should be checked against the course's guidance before release. **Fit Criterion:** A new account cannot be created without acknowledging the injury warning and the terms of use, and both can be reached from the settings screen at any time. **Hazard Analysis:** H4.2 **Related Requirements:** FUNC-01

ID	Requirement	Specification
PR-SC-06	Safe goal progression	**Description:** Goals that the product raises automatically shall increase gradually and include regular rest days. When a goal is missed MissedGoalLimit times in a row, the product shall suggest lowering it. Users shall be able to lower or override any goal themselves. **Rationale:** Goals that rise too quickly push users to overexert themselves, while goals that are missed again and again lead to shame and users giving up on exercise. **Fit Criterion:** Automatic goals never rise by more than MaxGoalIncrease per week, every goal plan includes at least one rest day per week, a lower goal is suggested after MissedGoalLimit misses in a row, and a user can lower any goal from the goal screen. **Hazard Analysis:** SR6 (H4.1, H4.2) **Related Requirements:** FUNC-04

ID	Requirement	Specification
PR-SC-07	Notification limits	**Description:** Users shall be able to set quiet hours during which no notifications are sent. By default, the product shall send no more than MaxDailyNotifications goal-related notifications per day. The wording of these messages is specified in UHR-POL-03, and which notifications a user receives is set under UHR-PER-02. **Rationale:** Too many notifications, or notifications at the wrong time, harm users' mental health and encourage compulsive use of the app. **Fit Criterion:** In testing, no notifications are sent during the user's quiet hours, and no more than MaxDailyNotifications goal-related notifications are sent in a day unless the user raises the limit. **Hazard Analysis:** SR7 (H4.1, H10.1) **Related Requirements:** UHR-PER-02, UHR-POL-02, UHR-POL-03

ID	Requirement	Specification
PR-SC-08	Safe leaderboard rules	**Description:** Leaderboards shall never rank users by absolute performance and shall never award currency or other rewards. Leaderboard participation is off by default (SEC-PRV-02). **Rationale:** Ranking users by absolute performance causes shame and anxiety for those who rank poorly. Rewarding leaderboard positions would also push users to overexert themselves to climb the rankings. **Fit Criterion:** Rankings are computed from each user's percentage of their own or the group's agreed goal, users who leave a group stop appearing on its leaderboard, and no currency is credited because of a ranking. **Hazard Analysis:** SR9 (H4.1, H8.1) **Related Requirements:** FUNC-13, FUNC-15, SEC-PRV-02

12.3 Precision or Accuracy Requirements

ID	Requirement	Specification
PR-PA-01	Exact money values	**Description:** Currency balances, prices, and trade amounts shall be exact to the cent, with no rounding error building up across operations, and shall be shown to two decimal places. **Rationale:** Floating-point rounding errors build up over many trades and would show users balances that do not add up. **Fit Criterion:** After a scripted sequence of 1,000 trades, the stored balance matches the expected balance exactly. **Related Requirements:** FUNC-07

ID	Requirement	Specification
PR-PA-02	Consistent currency awards	**Description:** The currency awarded for a set of activity data shall follow the published conversion rules exactly, and the same data shall always earn the same amount. **Rationale:** Users must be able to trust that effort is rewarded fairly, and the team must be able to check that it is. **Fit Criterion:** For a set of test activity records, the awarded currency matches the amount calculated by hand from the conversion rules. **Related Requirements:** FUNC-06
PR-PA-03	No double counting	**Description:** Activity reported by more than one source, such as the phone and a watch both counting the same steps, shall be counted only once. **Rationale:** Counting the same steps twice would over-reward users who own more devices, which the Hazard Analysis lists as a cause of poor learning. **Fit Criterion:** When the same walk is recorded by two sources, the currency earned equals the amount for one recording. **Related Requirements:** FUNC-03, FUNC-06, SEC-INT-01
PR-PA-04	Portfolio value	**Description:** The displayed portfolio value shall equal the sum of each holding's quantity times its current simulated price, plus uninvested currency. **Rationale:** The portfolio graph is the main tool for teaching compounding, so it has to be correct. **Fit Criterion:** For test portfolios, the displayed value matches the value calculated by hand to the cent. **Related Requirements:** FUNC-12

ID	Requirement	Specification
PR-PA-05	Workout dates	**Description:** Each workout shall be dated by the time it happened on the source device, not the time it was synced. When workout data arrives late, the affected streaks and rewards shall be recalculated. **Rationale:** Users who exercised on time should not lose their streak or currency because their device synced late or across a day boundary. **Fit Criterion:** Workouts synced after midnight or from a different time zone are recorded on the day they happened, and the affected streaks and rewards are updated once the late data arrives. **Hazard Analysis:** SR5 (H3.1, H11.1) **Related Requirements:** FUNC-03, FUNC-05, UHR-INT-01
PR-PA-06	Realistic market simulation	**Description:** The market simulation shall be calibrated against historical market data, so that its volatility and downturns are consistent with real markets. **Rationale:** Returns that are unrealistically smooth or high would teach users a false understanding of risk and compounding. **Fit Criterion:** Over a long simulated period, the average annual return and volatility are consistent with historical data for a broad market index, and downturns occur. **Hazard Analysis:** SR10 (H6.1) **Related Requirements:** FUNC-12

12.4 Robustness or Fault-Tolerance Requirements

ID	Requirement	Specification
PR-RFT-01	Offline activity	**Description:** Activity recorded while the phone is offline, or while the app is closed, shall be credited the next time the app syncs, for gaps of up to OfflineSyncWindow. **Rationale:** Users exercise in places without signal and will not open the app every day. Losing that activity would discourage them. **Fit Criterion:** Activity recorded during a gap of OfflineSyncWindow with no syncing is fully credited at the next sync. **Related Requirements:** FUNC-03

ID	Requirement	Specification
PR-RFT-02	Atomic writes	**Description:** Every operation that changes balances, holdings, rewards, or progress shall either complete fully or be rolled back. A trade shall update both the currency balance and the holdings, at the price the user confirmed, or not happen at all. **Rationale:** A write that fails halfway leaves balances, portfolios, or progress wrong. For example, a failed trade could take a user's currency without giving them the asset, or the other way around, and two simultaneous requests could produce duplicate rewards or a negative balance. **Fit Criterion:** Tests that simulate a failure at each step of a multi-step write leave the data unchanged. When the connection or server is cut during a test trade, the balance and holdings afterwards match either the state before the trade or the state after it, and completed trades execute at the confirmed price. A stress test of 100 concurrent trade requests against one account leaves every operation either fully completed or fully rolled back, with no negative balance. **Hazard Analysis:** SR12 (H7.2, H9.2) **Related Requirements:** FUNC-08, FUNC-09, FUNC-10

ID	Requirement	Specification
PR-RFT-03	Failing data sources	**Description:** If a fitness data source is unavailable or returns invalid data, the app shall keep working with its other sources and credit no currency for the invalid data. A failed workout sync shall be retried automatically, and if it still fails, the app shall tell the user which source has a problem and why. **Rationale:** Outages at Garmin or Health Connect are outside the team's control and should not make the whole app unusable. Users should find out about a failed sync from the app, not from a broken streak. **Fit Criterion:** With one source disabled or returning malformed data in testing, the app stays usable, retries the sync, then shows a message naming that source and the cause, and awards nothing for the bad data. **Hazard Analysis:** SR4 (H3.1) **Related Requirements:** FUNC-03, MAS-MNT-04
PR-RFT-04	Backend unavailable	**Description:** If the backend cannot be reached, the app shall show the last data it received, marked as possibly out of date, instead of crashing or showing empty screens. **Rationale:** Users on poor connections should still be able to see their progress. **Fit Criterion:** With the backend stopped, the app opens, shows the last known balance and portfolio, and displays an offline notice. **Related Requirements:** FUNC-07, FUNC-12, FUNC-16

12.5 Capacity Requirements

ID	Requirement	Specification
PR-C-01	Users	**Description:** The system shall support at least 500 registered users, with at least 100 using the app at the same time, without breaking the speed requirements above. **Rationale:** This covers the expected load during user testing, the final demo, and the Capstone EXPO, where many people may try the app at once. **Fit Criterion:** A load test with 100 simulated concurrent users meets PR-SL-02 and PR-SL-03. **Related Requirements:** PR-SL-01, PR-SL-02, PR-SL-03, PR-SE-03
PR-C-02	History	**Description:** The system shall store at least two years of activity and trade history for each user. **Rationale:** Compounding only shows up over long periods, so users need to be able to look back over their history. **Fit Criterion:** A test account loaded with two years of daily activity and weekly trades can display its full history, and its screens still meet PR-SL-02. **Related Requirements:** FUNC-11, FUNC-16, PR-SL-02

12.6 Scalability or Extensibility Requirements

ID	Requirement	Specification
PR-SE-01	Adding fitness sources	**Description:** A new fitness data source shall be addable without changing the currency conversion logic. **Rationale:** The team expects to add device integrations over time, and the Problem Statement lists integration with existing fitness technologies as a goal. **Fit Criterion:** Adding a new source only requires changes inside the integration module. **Related Requirements:** FUNC-03, MAS-ADPT-01
PR-SE-02	Adding asset types	**Description:** New types of assets shall be addable to the simulated market without changing the trading logic. **Rationale:** The product starts with stocks, bonds, and ETFs, and other asset types may be added later. **Fit Criterion:** Adding a new asset type only requires new asset definitions and market engine configuration, not changes to the buy and sell code. **Related Requirements:** FUNC-09, MAS-MNT-02
PR-SE-03	More server capacity	**Description:** The system shall be able to serve more users than PR-C-01 by adding server capacity, without any change to the mobile app. **Rationale:** If the product continues after the capstone, as the business plan explores, it will need to serve more users than PR-C-01. **Fit Criterion:** A load test at twice the PR-C-01 load, run after adding server capacity, meets PR-SL-02 and PR-SL-03 with the same version of the mobile app. **Related Requirements:** PR-SL-02, PR-SL-03, PR-C-01

12.7 Longevity Requirements

ID	Requirement	Specification
PR-L-01	Supported Android versions	**Description:** The app shall target the Android version that Google Play requires for new apps at the time of release, and keep working on Android versions released through the end of the capstone in April 2027. **Rationale:** Google Play stops accepting apps that target old Android versions, and the app must stay usable for the final demo and the EXPO. **Fit Criterion:** The release build meets Google Play's target version requirement and passes the test suite on the newest Android version available in April 2027. **Related Requirements:** LF-APP-03, PROD-02, COMP-STD-01
PR-L-02	Keeping user history	**Description:** Activity and portfolio history shall be kept for as long as the user's account exists, unless the user deletes it. **Rationale:** The product teaches that habits pay off over years. Deleting old history would remove the evidence of that. **Fit Criterion:** The system has no automatic deletion of history for active accounts. **Related Requirements:** FUNC-16

13 Operational and Environmental Requirements

13.1 Expected Physical Environment

The application will be operable through Android smartphones with cellular/Wi-Fi connectivity. Tablets are not supported. Inputs will be completed through

health tracking services that will be provided by the smart device that is running the software, or optionally received through other technologies used for health tracking.

13.2 Wider Environment Requirements

The app will be designed for the Android operating system for the Capstone Project. We must adhere to regulations/policies and limitations of the system. Software distribution and update management will be handled through the Google Play Store (SC-06).

13.3 Requirements for Interfacing with Adjacent Systems

Many outside systems will be used:

- Hypertext Transfer Protocol Secure (HTTPS) connection to external APIs
    1. External Technology Connection (Garmin Connect IQ SDK)
    2. Software-specific API for health-related tracking (Android Health Connect)
- HTTPS connection to internal APIs
    1. Java SpringBoot API for storing user data and retrieving from the database
- Database Connection
    1. McMaster University's MySQL server

13.4 Productization Requirements

ID	Requirement	Specification
PROD-01	Build and Quality Assurance	**Description:** The application will be subject to all the basic coding tests and standards needed to be published on the Google Play Store. **Rationale:** An app must be whole and completely working to provide value to the user base. **Fit Criterion:** All automated and manual written code tests must pass (100%) to be deemed fully complete. **Related Requirements:** MAS-MNT-03, COMP-STD-06, SC-06
PROD-02	Installability	**Description:** The application shall install from the Google Play Store onto any supported Android device without any setup beyond granting the permissions it asks for. **Rationale:** Users should be able to get started without technical help. Health Connect is built into recent versions of Android but must be installed separately on older ones, so the app has to guide users through that case. **Fit Criterion:** On each supported test device (see LF-APP-03), installing from Google Play completes in under 2 minutes on Wi-Fi and the app opens to the sign-up screen on first launch. Each permission is requested only when the feature that needs it is first used. If Health Connect is missing, the app directs the user to install it. Uninstalling removes all app data stored on the device. **Related Requirements:** FUNC-01, LF-APP-03, PR-L-01, COMP-STD-01

13.5 Release Requirements

The project releases will be aligned with the deliverables of the capstone project schedule:

1. Design Document Revision -1 and Fall Term User Interview report (CD-Fall) in Week 10.
2. Proof of Concept demonstration build in Weeks 11–12.
3. Revision 0 demonstration build in Weeks 18–19.
4. Final demonstration build in Week 23.
5. Final Documents (Revision 1), EXPO poster, and video in Week 25.
6. Capstone EXPO product and demonstration in Week 26.

14 Maintainability and Support Requirements

This section outlines the requirements for maintaining and supporting the software product throughout its lifecycle. It includes specifications for maintenance procedures, support services, and any necessary adaptability procedures to ensure the product remains functional and up-to-date.

14.1 Maintenance Requirements

ID	Requirement	Specification
MAS-MNT-01	Fitness Activity Addition Effort	**Description:** The system shall support the addition of new reward-eligible fitness activity types with minimal maintenance effort. **Rationale:** Future releases may introduce additional forms of exercise and adaptive fitness activities. **Fit Criterion:** A new activity type shall be implemented, deployed, and fully tested within 2 months. **Related Requirements:** FUNC-03, UHR-ACC-06

ID	Requirement	Specification
MAS-MNT-02	Investment Asset Addition Effort	**Description:** The system shall support the addition of new simulated investment assets with minimal maintenance effort. **Rationale:** Future versions of the application may expand the types of simulated investments available to users. **Fit Criterion:** A new investment asset shall be implemented, deployed, and fully tested within 2 months. **Related Requirements:** PR-SE-02
MAS-MNT-03	Corrective Maintenance Effort	**Description:** The system shall efficiently handle and correct software defects. **Rationale:** Timely defect resolution improves reliability and user satisfaction. **Fit Criterion:** Critical production defects shall be fixed and deployed within 5 business days of the defect being reported. **Related Requirements:** PROD-01, MAS-SUP-01
MAS-MNT-04	Third-Party Integration Maintenance	**Description:** The system shall support updates to external fitness platform integrations with minimal impact on overall system functionality. **Rationale:** External APIs may change independently of the application and require maintenance. **Fit Criterion:** Changes required to maintain compatibility with a supported fitness provider shall be deployed within 5 business days of identifying the incompatibility. **Related Requirements:** FUNC-03, PR-RFT-03, MAS-ADPT-01

ID	Requirement	Specification
MAS-MNT-05	Traceability	**Description:** The system shall provide sufficient documentation and traceability to allow future developers to efficiently understand previous changes and maintenance activities. **Rationale:** Accurate and up-to-date documentation improves maintainability by allowing future developers to understand previous modifications, known issues, and the reasoning behind architectural and implementation decisions. **Fit Criterion:** For every production release, developers shall be able to identify the implemented changes, associated issues, and documented design decisions within 30 minutes of searching project records. **Related Requirements:** COMP-LEG-07, DOC-04

14.2 Supportability Requirements

ID	Requirement	Specification
MAS-SUP-01	Contact Support Capability	**Description:** The system shall provide users with a mechanism to submit feedback or support requests for more complicated issues. **Rationale:** Some issues may require assistance beyond self-service documentation. **Fit Criterion:** Users shall be able to submit feedback or support requests from within the application at any time. These requests must be processed within 2 business days with a response sent back to the user regarding details pertaining to the plans and timeline to address the issue. **Related Requirements:** UHR-EOU-04, MAS-MNT-03, DOC-03

14.3 Adaptability Requirements

ID	Requirement	Specification
MAS-ADPT-01	External Fitness Provider Adaptability	**Description:** The system shall support the addition of new fitness data providers without requiring major changes to user workflows. **Rationale:** The ecosystem of fitness tracking applications and devices will evolve over time and additional providers may become desirable. **Fit Criterion:** Users shall be able to successfully connect newly supported fitness providers using the same workflow used for existing providers. **Related Requirements:** PR-SE-01, MAS-MNT-04, DOC-02

ID	Requirement	Specification
MAS-ADPT-02	Mobile Platform Portability	**Description:** The system shall support future deployment on additional mobile platforms beyond the initially targeted platform, Android. **Rationale:** Initial deployment is limited to Android (SC-01), but future releases may target additional mobile operating systems, such as iOS. **Fit Criterion:** A test client running outside Android can sign in to an existing account and retrieve the user's full account, activity, and portfolio data, with no migration step. **Related Requirements:** SC-01
MAS-ADPT-03	Future Artificial Intelligence Service Adaptability	**Description:** The system shall support future integration of educational content assisted by AI. **Rationale:** With the rapid growth of AI technologies, future releases may provide personalized fitness and financial learning experiences using AI technologies. **Fit Criterion:** AI-powered educational services shall be deployable without requiring modifications to existing user accounts. **Related Requirements:** COMP-LEG-05

15 Security Requirements

15.1 Access Requirements

ID	Requirement	Specification
SEC-ACC-01	Authenticated Access	**Description:** Access granted when a user signs in (FUNC-02) shall be checked on every request for personal fitness data, portfolios, or transactions, and shall expire after AccessExpiry. **Rationale:** Authentication protects personal data and virtual assets from unauthorized access. Expiring access limits the damage if a device or access credential is stolen. **Fit Criterion:** Every protected request with missing, invalid, or expired access is rejected with Hypertext Transfer Protocol (HTTP) status 401. After AccessExpiry, the app must refresh its access or the user must sign in again. **Related Requirements:** FUNC-02, COMP-LEG-02, COMP-STD-07
SEC-ACC-02	Role-Based Authorization	**Description:** Role-based access control shall restrict users to their own records and limit administrative or evaluator access to authorized functions. **Rationale:** Access restrictions prevent unauthorized modifications and cross-account data exposure. **Fit Criterion:** JUnit security tests shall confirm that attempts by standard users to access or modify another user's records return HTTP 403 in all tested cases. **Hazard Analysis:** SR3 (H2.1, H8.1, H9.1) **Related Requirements:** FUNC-02, SEC-PRV-02

ID	Requirement	Specification
SEC-ACC-03	Fitness Integration Authorization	**Description:** Flexin' shall request only necessary read permissions for supported integrations, Android Health Connect and Garmin. Integration credentials shall be stored encrypted on the device, where other apps cannot read them. **Rationale:** Secure credential handling protects external accounts and limits access to fitness data. **Fit Criterion:** A security review confirms that integration tokens are stored only in encrypted device storage that other apps cannot read, and finds no credentials, tokens, or secrets in the project repository. **Related Requirements:** FUNC-03

ID	Requirement	Specification
SEC-ACC-04	Password and Sign-In Protection	**Description:** Flexin' shall enforce a minimum password strength, offer optional multi-factor authentication through the user's email, and temporarily lock sign-in after repeated failed attempts. Users shall be able to reset their password through email. **Rationale:** Weak passwords and unlimited sign-in attempts let attackers reach users' private data, while legitimate users must still be able to regain access to their accounts. **Fit Criterion:** Passwords shorter than MinPasswordLength characters or found on a common-password list shall be rejected. Sign-in to an account shall be locked for LockoutDuration after LockoutAttempts consecutive failed attempts. A user who enables multi-factor authentication shall be asked for an email code at each new sign-in, and a password reset email shall let a locked-out user sign in again. **Hazard Analysis:** SR1 (H1.1, H1.2) **Related Requirements:** FUNC-01, FUNC-02, SEC-AUD-02, SEC-IMM-02

ID	Requirement	Specification
SEC-ACC-05	Session End and Device Unlinking	**Description:** Flexin' shall end a user's session after a period of inactivity or when the user signs out, and shall unlink the user's fitness devices on sign-out. **Rationale:** Sessions and device links left open on shared or lost devices let other people access an account or send their health data to it. **Fit Criterion:** A session unused for SessionTimeout shall require the user to sign in again. After sign-out, the user's access and refresh tokens shall be rejected with HTTP 401, and no integration from that device shall be able to submit activity to the account. **Hazard Analysis:** SR2 (H1.2, H11.2) **Related Requirements:** FUNC-02

15.2 Integrity Requirements

ID	Requirement	Specification
SEC-INT-01	Activity Validation	**Description:** The system shall validate activity metrics and detect overlapping timestamps and implausible activity values before awarding virtual currency. Activity reported more than once is handled by PR-PA-03. **Rationale:** Validation prevents activity manipulation and unauthorized currency generation. **Fit Criterion:** Automated tests shall confirm that duplicate or flagged implausible activity records are marked unverified and receive zero virtual currency. Sustained cadence above MaxCadence shall be flagged for verification. **Related Requirements:** FUNC-03, FUNC-06, PR-PA-03, SEC-AUD-02, SEC-IMM-03
SEC-INT-02	Server-Side Valuation	**Description:** Asset prices, portfolio valuations, and currency amounts shall be calculated by the system's servers. Values sent by the mobile app shall never be accepted as authoritative. **Rationale:** Server-side calculations prevent client manipulation of prices and balances. **Fit Criterion:** Trade requests attempting to specify unauthorized execution prices shall be rejected with HTTP 400. **Related Requirements:** FUNC-09, FUNC-10

ID	Requirement	Specification
SEC-INT-03	Data Backups and Recovery	**Description:** Flexin' shall keep regular backups of all user data, including accounts, activity history, balances, and portfolios, and shall be able to restore that data after data corruption or an infrastructure failure. **Rationale:** Without backups, a database failure or bad write would leave users' balances, portfolios, and progress permanently lost. Fast recovery limits how long users are affected. **Fit Criterion:** The production database is backed up at least once every BackupInterval, and each backup is kept for BackupRetention. Data can be restored from the most recent backup within RestoreTime, and a restore into a test environment is performed successfully before each release. **Hazard Analysis:** SR12 (H7.2, H9.2) **Related Requirements:** SEC-PRV-03

15.3 Privacy Requirements

ID	Requirement	Specification
SEC-PRV-01	Health Data Protection	**Description:** Sensitive personal information and fitness telemetry shall be protected through encryption in transit and at rest. **Rationale:** Encryption protects confidential user information against unauthorized disclosure. **Fit Criterion:** Client-server communications shall use Transport Layer Security (TLS) 1.3. Stored health and biometric data shall be encrypted at rest using the Advanced Encryption Standard (AES) with 256-bit keys, or stronger. **Related Requirements:** COMP-LEG-02, COMP-STD-07

ID	Requirement	Specification
SEC-PRV-02	Private by Default	**Description:** All profile and health data shall be private by default. A field shall be shared with other users only after the user opts in, and backend access rules shall deny access to any data that has not been shared. Leaderboard participation shall be opt-in, and users may share activity streaks and simulated portfolio returns using a chosen alias. **Rationale:** Sensitive information such as weight, goals, and missed workouts must never be exposed to other users or unauthorized parties without consent. Users should control whether their activity and simulated financial performance are visible to others. **Fit Criterion:** All sharing settings, including leaderboard sharing, shall be disabled for new accounts. Enabling sharing shall require explicit consent and disclose which pseudonymous information and statistics will be shared. Automated access-rule tests run in the CI/CD pipeline shall confirm before each release that no user can read another user's unshared data. **Hazard Analysis:** SR3 (H2.1, H8.1, H9.1), SR9 (H4.1, H8.1) **Related Requirements:** FUNC-13, PR-SC-08, SEC-ACC-02, COMP-LEG-02

ID	Requirement	Specification
SEC-PRV-03	User Data Deletion	**Description:** Users shall be able to delete their account and associated personal information at any time from the application's settings menu. **Rationale:** Users should retain control over their personal information and account history. **Fit Criterion:** After a deletion request, personal health metrics, credentials, and fitness history are removed from the active database within DeletionWindow. Because data is backed up by default (SEC-INT-03), copies held in backups are removed as those backups expire, within BackupRetention. Audit records (SEC-AUD-01) keep only an anonymized account identifier that can no longer be linked to the deleted user. **Related Requirements:** SEC-INT-03, SEC-AUD-01, COMP-LEG-04

15.4 Audit Requirements

ID	Requirement	Specification
SEC-AUD-01	Economic Event Logging	**Description:** Flexin' shall maintain an append-only audit ledger for currency rewards, simulated trades, dividends, and market price updates. **Rationale:** Auditable records support troubleshooting, transaction verification, and evaluation of the simulated economy. **Fit Criterion:** Database permissions shall prevent UPDATE and DELETE operations on audit records. Every recorded transaction shall reference a valid source activity or market event, where applicable. **Related Requirements:** FUNC-06, FUNC-11, SEC-PRV-03
SEC-AUD-02	Security Event Logging	**Description:** The backend shall log failed authentication, token validation failures, duplicate activity submissions, anti-exploit flags, and runtime exceptions. **Rationale:** Security logs help identify suspicious activity and diagnose integration or system failures. **Fit Criterion:** Structured logs shall capture severity, client Internet Protocol (IP) address where available, authenticated user ID where applicable, and exception details for runtime errors. **Related Requirements:** SEC-ACC-04, SEC-INT-01, SEC-IMM-02

15.5 Immunity Requirements

ID	Requirement	Specification
SEC-IMM-01	Injection Defense	**Description:** The system shall be protected against Structured Query Language (SQL) injection, cross-site scripting (XSS), and command injection. **Rationale:** Injection vulnerabilities could compromise database integrity and expose user information. **Fit Criterion:** Before each release, an automated security scan and a code review find zero unresolved high-severity injection vulnerabilities. **Related Requirements:** COMP-STD-07
SEC-IMM-02	Rate Limiting	**Description:** The system shall limit request rates for authentication, trade submission, and activity synchronization endpoints. **Rationale:** Rate limiting reduces brute-force attempts, automated abuse, and service overload. **Fit Criterion:** A limit of RateLimit per authenticated user or client IP address shall be enforced. Excess requests shall receive HTTP 429. Automated tests shall verify enforcement. **Related Requirements:** FUNC-02, FUNC-09, SEC-ACC-04, SEC-AUD-02

ID	Requirement	Specification
SEC-IMM-03	Client Environment Validation	**Description:** The Android client shall detect supported indicators of rooted or otherwise untrusted environments and restrict sensitive functionality when tampering is detected outside developer builds. **Rationale:** Compromised environments can increase the risk of credential exposure and activity manipulation. **Fit Criterion:** In supported production configurations, detected root or tampering indicators shall trigger restrictions on community leaderboard access and virtual currency minting. Developer builds shall remain available for authorized testing. **Related Requirements:** FUNC-06, FUNC-13, SEC-INT-01

16 Cultural Requirements

This section outlines the cultural requirements for the software system. Cultural requirements refer to the expectations and norms of the target user base, which may vary based on geographic location, language, social norms, and other cultural factors. These requirements ensure that the software is culturally sensitive and appropriate for its intended audience.

16.1 Cultural Requirements

ID	Requirement	Specification
CUL-01	Supports Canadian Languages	**Description:** The system shall present all user-facing functionality with the option to choose between English and French. **Rationale:** The application will initially be deployed strictly within Canada, where English and French are the country's official languages. **Fit Criterion:** Every screen can be shown in English or French, chosen in settings, with no untranslated text. **Related Requirements:** UHR-INT-01, UHR-UND-01
CUL-02	Culturally Neutral Educational Content	**Description:** The system shall present fitness and financial education content using culturally neutral language and examples (i.e. sample companies in the virtual market). **Rationale:** The Canadian population consists of individuals from diverse cultural, religious, and socioeconomic backgrounds. Educational content should be inclusive and broadly applicable. **Fit Criterion:** Educational content shall not require familiarity with a specific cultural background, religion, ethnicity, or region to be understood. **Related Requirements:** UHR-LRN-04, UHR-UND-01, UHR-POL-01

17 Compliance Requirements

This section outlines the legal and regulatory requirements that the system must adhere to, ensuring that it operates within the bounds of applicable laws and standards. Compliance with these requirements is essential for the system's legitimacy, user trust, and overall success.

17.1 Legal Requirements

The financial advice disclaimer and the exercise risk acknowledgement are legal requirements as well as safety requirements. To keep each requirement in one place, they are specified only in PR-SC-04 and PR-SC-05 (Safety-Critical Requirements).

ID	Requirement	Specification
COMP-LEG-01	User Consent for Health Data Collection	**Description:** The system shall obtain explicit user consent before collecting, storing, or processing any health or fitness related data from external devices or applications. **Rationale:** The system requires the collection of sensitive health and fitness data in accordance with requirement FUNC-03. Obtaining explicit user consent ensures compliance with PIPEDA and the Personal Health Information Protection Act (PHIPA). **Fit Criterion:** No fitness data shall be retrieved until the user has completed the system's permission and consent process. **Related Requirements:** FUNC-03

ID	Requirement	Specification
COMP-LEG-02	Protection of Personal Information	**Description:** The collection, use, storage, and disclosure of personally identifiable information (PII), beyond just fitness data, shall follow the fair information principles of PIPEDA, and of PHIPA if it applies (see Open Issues). **Rationale:** The system stores personal information including name, height, weight, health conditions, and financial learning metrics/goals. Protecting this information is necessary to adhere to mandatory privacy regulations and maintain trust. This also helps mitigate the risks H2.1 and H11.2 identified in the hazard analysis. **Fit Criterion:** Before each release, a privacy review checks the product against each of PIPEDA's ten fair information principles and records how each one is met, with none left unaddressed. The technical protections the review relies on are specified in SEC-ACC-01, SEC-PRV-01, and SEC-PRV-02. **Related Requirements:** SEC-ACC-01, SEC-PRV-01, SEC-PRV-02, COMP-LEG-04

ID	Requirement	Specification
COMP-LEG-03	Sharing of User Data	**Description:** The system shall not share **any** user data with third parties without explicit user consent, in accordance with PIPEDA. **Rationale:** The system collects sensitive health, financial and personal information, and the system will be involved with other third party systems (for managing related health data - FUNC-03). If sponsored rewards (FUNC-17, a stretch goal) are added later, sponsors will also be third parties covered by this requirement. Sharing this data without consent would violate user privacy and the regulations outlined in PIPEDA. **Fit Criterion:** All data will remain securely stored and not shared with third parties **under any circumstances** without the user explicitly completing the user permission and consent process. **Related Requirements:** FUNC-03, FUNC-17

ID	Requirement	Specification
COMP-LEG-04	User Data Deletion Rights	**Description:** Account deletion, as specified in SEC-PRV-03, shall meet PIPEDA's requirements for withdrawing consent and disposing of personal information. **Rationale:** Users must have control over their personal information to comply with PIPEDA regarding privacy regulations and PPCDA regarding data ownership and the user's right to disposal. **Fit Criterion:** The privacy review in COMP-LEG-02 confirms that account deletion under SEC-PRV-03 meets these requirements. **Related Requirements:** SEC-PRV-03, COMP-LEG-02
COMP-LEG-05	AI-Content Disclosure	**Description:** The system shall disclose when educational content is generated or assisted by an artificial intelligence system. **Rationale:** With the potential for AI to aid in the virtual market simulation, users should be informed when content is generated by AI to promote transparency and allow them to appropriately evaluate the information received. **Fit Criterion:** All AI-generated educational content shall include a visible notification indicating that the content was generated or assisted by AI. Additionally, the settings menu will contain an information section that explicitly discloses all components generated with the assistance of AI. **Related Requirements:** UHR-LRN-04, MAS-ADPT-03

ID	Requirement	Specification
COMP-LEG-06	Age Verification	**Description:** The system shall verify a user's age to be a legal adult (MinAge or older) during account creation. **Rationale:** The system collects personal and health-related information that may be subject to additional restrictions for minors. It is important that users are of age to provide informed consent and understand the implications of sharing their personal information. **Fit Criterion:** During onboarding, users shall validate their age before account activation and their accounts will not be activated if they are not of legal age (MinAge or older). **Related Requirements:** FUNC-01
COMP-LEG-07	Third-Party Licensing Compliance	**Description:** The system shall comply with all licensing requirements associated with third-party libraries, APIs, SDKs, and external content used by the application. **Rationale:** The application relies on external technologies and services whose licenses impose legal obligations on software developers during the development of this system. **Fit Criterion:** All third-party components included in a release shall have documented license reviews and approval prior to deployment. **Related Requirements:** MAS-MNT-05, SC-02, SC-03, SC-04

17.2 Standards Compliance Requirements

ID	Requirement	Specification
COMP-STD-01	Android Platform Compliance	**Description:** The system shall comply with the Google Play Developer Program Policies. **Rationale:** The application is intended to be deployed on the Google Play Store as to be accessible on Android devices and as a result must conform to platform requirements. **Fit Criterion:** The application shall successfully pass all mandatory checks required for distribution through the Google Play Store. **Related Requirements:** PR-L-01, PROD-02, SC-06
COMP-STD-02	Accessibility Compliance	**Description:** The system shall comply with WCAG 2.1 Level AA for user-facing functionality. **Rationale:** The system should offer a simple and delightful user experience that should be accessible to users with varying abilities (i.e. users with colour-blindness, hearing impairments, etc.). **Fit Criterion:** All major user workflows shall satisfy WCAG 2.1 Level AA success criteria as evaluated using automated accessibility tools and manual testing. **Related Requirements:** LF-APP-04, UHR-ACC-01, UHR-ACC-02, UHR-ACC-03, UHR-ACC-04, UHR-ACC-05

ID	Requirement	Specification
COMP-STD-03	TypeScript Coding Standards Compliance	**Description:** The frontend codebase shall comply with the Google TypeScript Style Guide. **Rationale:** Consistent coding standards improve maintainability, readability, and long-term project quality. **Fit Criterion:** All pull requests shall pass configured linting and formatting checks before merging into the main branch. **Related Requirements:** SC-02
COMP-STD-04	Java Coding Standards Compliance	**Description:** The backend codebase shall comply with the Oracle Java Code Conventions. **Rationale:** Standardized coding practices improve maintainability, code review effectiveness, and software quality. **Fit Criterion:** All pull requests shall pass configured linting and formatting checks before merging into the main branch. **Related Requirements:** SC-03
COMP-STD-05	Database Compliance	**Description:** The system shall store data in compliance with the MySQL Reference Manual and the SQL Standard (International Organization for Standardization and International Electrotechnical Commission (ISO/IEC) 9075). **Rationale:** The project architecture relies upon MySQL as the primary relational database system. **Fit Criterion:** All schema definitions and queries will be tested to execute successfully on the supported MySQL version without compatibility warnings or errors. **Related Requirements:** SC-03

ID	Requirement	Specification
COMP-STD-06	Software Testing Standard Compliance	**Description:** The system shall undergo thorough and complete testing in accordance with the Institute of Electrical and Electronics Engineers (IEEE) 29119 Software Testing Standard. **Rationale:** Consistent testing practices reduce defects and improve software reliability. The system will comply with this standard to provide a unified, process-driven framework for software testing. **Fit Criterion:** All code merged into the main branch shall pass automated test execution in the CI/CD pipeline based on tests that were developed in accordance with the IEEE 29119 Software Testing Standard. **Related Requirements:** PROD-01
COMP-STD-07	Information Security Compliance	**Description:** The system shall be designed in accordance with the security principles outlined in ISO/IEC 27001. **Rationale:** The application stores personal, financial-learning, and fitness-related information that must be protected against unauthorized access and disclosure. **Fit Criterion:** Before release, a checklist maps the security requirements to the applicable ISO/IEC 27001 Annex A controls, and every applicable control is either addressed or justified as out of scope. **Related Requirements:** SEC-ACC-01, SEC-PRV-01, SEC-IMM-01

18 Open Issues

The following issues were found while reviewing this document for consistency and have not yet been resolved.

1. **Business data model.** The Business Data Model figure has not yet been updated for the attributes added to the Data Dictionary (for example Email, Date of Birth, Goal Progress, and Transaction Type). The figure also shows a portfolio holding many assets, while the Portfolio table in the Data Dictionary has a single Asset ID and Quantity; a separate holding class is likely needed to reconcile the two.

2. **Claim Virtual Currency flowchart.** The flowchart for this use case (Figure 4) still shows users collecting currency by completing financial lessons. Currency is now earned only through physical activity, so the image needs to be updated.

3. **Privacy legislation.** The team needs to confirm whether PHIPA applies to Flexin', and what "PPCDA" in COMP-LEG-04 refers to.

4. **API hosting.** The database will be hosted on McMaster University's MySQL server, but where the Spring API will be hosted has not been decided.

5. **Other documents.** The Problem Statement describes users buying real financial assets at a simulated price, which differs from the fictional market decided on in this document.

19 Likely and Unlikely Changes

19.1 Likely Changes

LC-01 Symbolic constant values. The values in Table 2 marked "initial" will be tuned at the Currency Decision milestone and after user testing.

LC-02 Activity-to-currency conversion rules. How much currency each type of activity earns will be set at the Currency Decision milestone and is expected to be adjusted as testing shows how users respond.

LC-03 Supported fitness data sources. Health Connect and Garmin are supported first; more devices and services are likely to be added (PR-SE-01, MAS-ADPT-01).

LC-04 Asset types and market calibration. New fictional asset types may be added (PR-SE-02, MAS-MNT-02), and the market engine's calibration will be refined (PR-PA-06).

LC-05 User interface details. Screen layouts, navigation, wording, and educational content will change after each round of user testing.

LC-06 Notifications. The kinds of notifications and their default settings are likely to change based on user feedback.

LC-07 API hosting. Where the API is hosted is still open (see Open Issues).

LC-08 Additional platforms. An iOS version is a stretch goal (MAS-ADPT-02).

LC-09 Sponsored rewards. Redeeming currency for sponsor rewards may be added if partnerships are found (FUNC-17).

LC-10 AI-assisted content. Educational content assisted by AI may be added later (MAS-ADPT-03, COMP-LEG-05).

19.2 Unlikely Changes

UC-01 The core loop. Physical activity earns virtual currency that is invested in a simulated market. This is the product concept from the Problem Statement; changing it would make a different product.

UC-02 No real money. The product will not handle real money or real trades (SC-05). Doing so would bring financial regulation that is outside the project's scope and would expose users to real losses.

UC-03 Fictional market. Assets and prices are fictional and generated by the product's own engine, not live market data. This avoids paid data feeds (Budget Constraints) and makes it clear that nothing in the app is advice about real investments (PR-SC-04).

UC-04 Android first. The capstone release targets Android only (SC-01); this follows from the schedule and the team's scope decisions.

UC-05 Privacy and age obligations. Consent, data protection, deletion, and the minimum age (COMP-LEG-01 to COMP-LEG-06) come from law and will not be relaxed.

UC-06 Leaderboard rules. Leaderboards stay opt-in, ranked by progress towards goals, and reward-free (PR-SC-08), because they address hazards identified in the Hazard Analysis.

UC-07 Technology stack. The frontend and backend technologies (SC-02, SC-03) were agreed in the Development Plan and development has been planned around them.

20 Off-the-Shelf Solutions

20.1 Ready-Made Products

The team looked at existing products that could meet the project's goals without building something new. Each covers part of the problem, but none connects fitness habits to learning how investments grow.

Stock market simulators (e.g. Investopedia Simulator): Free simulators give users virtual cash to trade real securities at market prices, and some let users compete in private groups. They teach investing well, but only to people who are already motivated to learn about it. The Problem Statement identifies young adults who are not yet interested as the group furthest behind, so a simulator alone does not reach them.

Step reward apps (e.g. Sweatcoin): These apps turn steps into a virtual currency that can be spent on offers and products. They reward activity, but the currency is spent rather than invested, so they teach nothing about compounding.

Move-to-earn crypto apps (e.g. STEPN): These reward exercise with cryptocurrency tokens that have real market value. They link exercise to money, but expose users to real financial loss. STEPN's reward token lost most of its value within months in 2022. This approach conflicts with constraint SC-05 (simulated market only) and is not suitable.

Fitness trackers (e.g. Strava, Garmin Connect, Health Connect): These record activity accurately but do not offer any financial component. Flexin' will use Health Connect and Garmin as data sources rather than replace them.

Since no ready-made product covers both halves of the problem, the team will build Flexin', reusing the components and ideas listed below.

20.2 Reusable Components

The following existing components are candidates for reuse. Each must have a license compatible with the project's proprietary license, and must be free to use

within the project's budget.

Fitness data:

- `react-native-health-connect`, an open-source (Massachusetts Institute of Technology (MIT) License) React Native library for reading data from Android Health Connect.
- The Garmin Connect IQ SDK for Garmin watches.

Mobile client:

- React Navigation for moving between screens.
- An open-source React Native charting library for the portfolio graph and activity history. The specific library will be chosen during design.

Backend:

- Spring Boot to set up and run the API.
- Spring Security for authentication and access control, rather than writing account security from scratch.
- Spring Data Jakarta Persistence API (JPA) for reading and writing the MySQL database.

Historical market data: A freely available historical data set for a broad market index, used only to calibrate the market engine so that its fictional prices follow real-world trends (see PR-PA-06). The product does not use live prices.

Testing and tooling: JUnit, Detox, ESLint, and GitHub Actions, as listed in the Development Plan.

20.3 Products That Can Be Copied

The products below are not reused directly, but parts of their design can be adapted for Flexin'. Only ideas will be copied, not code, artwork, or other protected material.

Investment simulators: The layout of portfolio, trade, and order history screens, which users of investing apps will already recognize. Short explanations of financial terms placed next to where the terms are used.

Step reward apps: Converting steps into currency at a fixed rate, with a daily limit on how much can be earned (see PR-SC-01).

Strava: Leaderboards, clubs, and activity feeds that let friends follow each other's progress. These can be adapted for the community leaderboard and friend groups.

Habit and learning apps (e.g. Duolingo): Streaks, short daily goals, and reminders that build a daily habit. These must be adapted carefully so that missing a day is not punished (see PR-SC-02).

21 New Problems

N/A
Reason: Due to being in the early stages of the project, there are no development problems at this time. This section is subject to change as new problems arise in later revisions of this document.

21.1 Effects on the Current Environment

N/A

21.2 Effects on the Installed Systems

N/A

21.3 Potential User Problems

N/A

21.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

N/A

21.5 Follow-Up Problems

N/A

22 Tasks

This section outlines the tasks that will be undertaken during the development of the solution. Each task is associated with a specific duration, start and finish date, and any dependencies on other tasks. The tasks are designed to ensure that the project progresses in a structured manner, with clear milestones and deliverables. This section also includes the phase-in plan identifying when requirements will be phased into the development process.

22.1 Project Planning

The team will follow an agile development process throughout the development of this solution, with a focus on iterative development and continuous feedback. The project will be divided into several sprints, each lasting two weeks. At the end of each sprint, the team will conduct a review to assess progress and make necessary adjustments. Each sprint will consist of planning, developing, testing, and deploying. This will allow the team to make iterative progress while still ensuring that previous iterations are effective and meet the requirements of the stakeholders. The detailed breakdown of weekly tasks and deliverables can be found in section 9 of the development plan and is the basis for the following project task breakdown:

Task	**Duration**	**Start**	**Finish**	**Dependency**
Requirements Gathering & Analysis	4 weeks	21-Sep-26	18-Oct-26	None
User Interviews & Validation	3 weeks	05-Oct-26	25-Oct-26	Requirements Gathering & Analysis
Fitness API Research & Integration Design	3 weeks	12-Oct-26	01-Nov-26	Requirements Gathering & Analysis
Virtual Currency System Design	2 weeks	26-Oct-26	08-Nov-26	User Interviews & Validation
Market Simulation Design	3 weeks	19-Oct-26	08-Nov-26	Requirements Gathering & Analysis
UI and user experience (UX) Design & Prototyping	4 weeks	26-Oct-26	22-Nov-26	User Interviews & Validation

Task	Duration	Start	Finish	Dependency
Proof of Concept Development	5 weeks	02-Nov-26	06-Dec-26	Fitness API Design, Currency Design
Backend Architecture & Database Development	5 weeks	11-Jan-27	14-Feb-27	Design Document
Frontend Development	6 weeks	11-Jan-27	21-Feb-27	UI Design
Fitness Provider Integration	5 weeks	18-Jan-27	21-Feb-27	Backend Development
Portfolio Management Implementation	4 weeks	25-Jan-27	21-Feb-27	Backend Development
Gamification & Rewards Implementation	4 weeks	25-Jan-27	21-Feb-27	Currency Design
Rev 0 Demo Preparation	2 weeks	25-Jan-27	07-Feb-27	Core Workflow Demonstrable
User Testing Round 1	2 weeks	25-Jan-27	07-Feb-27	Demo Build for User Testing
Accessibility, Security, & Compliance Improvements	2 weeks	15-Feb-27	28-Feb-27	User Testing Round 1
Final User Testing	2 weeks	01-Mar-27	14-Mar-27	Accessibility, Security, & Compliance Improvements
V&V Execution	4 weeks	15-Feb-27	14-Mar-27	Most Development Tasks Complete
Final Bug Fixes & Stabilization	2 weeks	08-Mar-27	21-Mar-27	V&V Execution
Final Demo Preparation	1 week	15-Mar-27	21-Mar-27	Final Bug Fixes & Stabilization
EXPO Preparation	2 weeks	22-Mar-27	05-Apr-27	Final Demo Preparation

Table 6: Project Schedule and Task Breakdown

22.2 Planning of the Development Phases

The Flexin' project will be implemented incrementally through a series of development phases. Requirements are allocated to phases based on technical depen-

dencies, implementation risk, stakeholder priority, and project milestones. Earlier phases focus on establishing the core platform infrastructure and fitness-to-investment workflow, while later phases introduce engagement features, usability enhancements, accessibility improvements, and compliance requirements.

Each phase's deadline is the date its last task finishes in Table 6. Phases 1 to 3 are built in parallel between January 11 and February 21, 2027, so they share a deadline; their order is the order in which their features are started and prioritized, and the Rev 0 demo (February 1 to 12, 2027) shows them in progress. The requirements identified below represent the planned allocation at the time of writing. Requirement assignments and phase deadlines may be refined during development as new information becomes available through user testing, stakeholder feedback, and technical investigation.

22.2.1 Phase 1 — Foundation and Core Platform

Deadline: February 21, 2027

This phase establishes the core application infrastructure including account management, authentication, fitness integration, virtual currency generation, and synchronization with supported health platforms. The primary objective is to demonstrate the end-to-end workflow of completing fitness activity and receiving virtual currency rewards.

Priority Requirements: FUNC-01 through FUNC-07, SEC-ACC-01, SEC-ACC-03, SEC-PRV-01, PR-SL-01, PR-SL-02, PR-SL-04, PR-SC-01, PR-SC-04, PR-SC-05, PR-PA-02, PR-PA-03, PR-PA-05, PR-RFT-01, PR-RFT-03, PR-SE-01, SEC-ACC-02, UHR-PER-01, UHR-INT-01

22.2.2 Phase 2 — Simulated Investing

Deadline: February 21, 2027

This phase introduces the simulated financial market and portfolio management capabilities. Users will be able to purchase and sell investment assets, manage portfolios, and view investment performance using virtual currency earned through physical activity.

Priority Requirements: FUNC-08 through FUNC-12, PR-PA-01, PR-PA-04, PR-PA-06, SEC-INT-01, SEC-INT-02, SEC-AUD-01, PR-SL-03, PR-SC-03, PR-RFT-02, PR-SE-02

22.2.3 Phase 3 — Community and Gamification

Deadline: February 21, 2027

This phase focuses on user engagement features including leaderboards, friend groups, private competitions, and progress reporting. Sponsor-based reward redemption (FUNC-17) is a stretch goal and is not scheduled (see the Waiting Room).

Priority Requirements: FUNC-13 through FUNC-16, UHR-PER-03, UHR-PER-04, UHR-POL-04, UHR-POL-05, UHR-PER-02, PR-SC-02, PR-SC-06 through PR-SC-08, SEC-PRV-02

22.2.4 Phase 4 — Usability, Accessibility, and Learning

Deadline: February 28, 2027

This phase refines the overall user experience through interface improvements, educational content, accessibility enhancements, personalization features, and user feedback gathered through testing activities.

Priority Requirements: LF-APP-01 through LF-APP-05, LF-STY-01 through LF-STY-04, UHR-EOU-01 through UHR-EOU-09, UHR-LRN-01 through UHR-LRN-04, UHR-UND-01 through UHR-UND-03, UHR-ACC-01 through UHR-ACC-06, UHR-POL-01 through UHR-POL-03, CUL-01, CUL-02, PR-RFT-04

22.2.5 Phase 5 — Security, Compliance, and Release Preparation

Deadline: March 14, 2027

This phase focuses on hardening the application prior to release. Security controls, privacy requirements, documentation, training materials, legal compliance requirements, maintenance support, and final validation activities are completed during this phase.

Priority Requirements: COMP-LEG-01 through COMP-LEG-07, COMP-STD-01 through COMP-STD-07, SEC-ACC-04, SEC-ACC-05, SEC-IMM-01 through SEC-IMM-03, MAS-MNT-01 through MAS-MNT-05, MAS-SUP-01, MAS-ADPT-01 through MAS-ADPT-03, DOC-01 through DOC-04, TRN-01 through TRN-03, SEC-INT-03, SEC-PRV-03, SEC-AUD-02, PROD-01, PROD-02, PR-C-01, PR-C-02, PR-SE-03, PR-L-01, PR-L-02

23 Migration to the New Product

N/A
Reason: This system is being created as a new concept from scratch. No existing system exists that will be replaced by this new solution.

23.1 Requirements for Migration to the New Product

N/A

23.2 Data That Has to be Modified or Translated for the New System

N/A

24 Costs

N/A
Reason: Currently there are no costs associated with the development of this system other than team members' time. This section is subject to change as development costs arise in later revisions of this document.

25 User Documentation and Training

25.1 User Documentation Requirements

ID	Requirement	Specification
DOC-01	Financial and Fitness Glossary	**Description:** Flexin' shall provide a searchable in-app glossary of the terms explained under UHR-LRN-04, including compound interest, inflation, ETFs, bond yields, and portfolio diversification. **Rationale:** A single list of terms lets novice users review concepts they have met in the app, separately from the screen where they first saw them. **Fit Criterion:** Every term that has a contextual explanation under UHR-LRN-04 also appears in the glossary. At least 90% of novice test participants locate and understand an ETF or compound return definition within 15 seconds. **Related Requirements:** UHR-LRN-02, UHR-LRN-04
DOC-02	Fitness Integration Guide	**Description:** Flexin' shall provide an illustrated setup guide explaining how to authorize and configure supported fitness integrations and troubleshoot synchronization issues. **Rationale:** Clear instructions help users connect their fitness data sources successfully. **Fit Criterion:** Screenshot-assisted instructions shall cover all supported providers. At least 95% of test participants shall complete authorization using the guide alone. **Related Requirements:** FUNC-03, UHR-EOU-06, MAS-ADPT-01

ID	Requirement	Specification
DOC-03	Searchable Help Center	**Description:** Flexin' shall provide an in-app, searchable help center, reachable from the settings menu, that explains all major features and covers account management and recovery, activity synchronization, activity rewards, portfolio management, simulated market mechanics, virtual currency disclaimers, and health-data privacy. **Rationale:** Users should be able to resolve common questions themselves without contacting the team. Accessible documentation also improves transparency and builds user trust. **Fit Criterion:** The help center covers every topic listed above, can be opened from the settings menu at any time, works without an external browser, supports real-time keyword filtering, and returns search results in under 200 milliseconds. **Related Requirements:** MAS-SUP-01
DOC-04	Digital Manual and Release Notes	**Description:** The team shall maintain a digital user manual and release notes in the project GitHub documentation repository. **Rationale:** Documentation enables supervisors, reviewers, and evaluators to verify functionality and assess project deliverables. **Fit Criterion:** A self-contained Portable Document Format (PDF) and Markdown manual shall document architecture, navigation, data flows, and troubleshooting, with release notes for major project milestones. **Related Requirements:** MAS-MNT-05, TRN-03

25.2 Training Requirements

ID	Requirement	Specification
TRN-01	Interactive Onboarding	**Description:** After account creation, Flexin' shall offer an optional, dismissible four-step walkthrough covering fitness goals, activity rewards, simulated investing, and the compounding chart. **Rationale:** Guided interaction introduces the application's core features through practical examples. **Fit Criterion:** The walkthrough shall take under 90 seconds, provide a visible "Skip Tour" option on every step, and remain accessible through settings. **Related Requirements:** UHR-EOU-08, UHR-LRN-01
TRN-02	Progressive Feature Discovery	**Description:** Flexin' shall progressively introduce advanced features, including portfolio rebalancing, volatility filters, and leaderboard sharing, as users gain experience. **Rationale:** Gradual feature discovery reduces cognitive overload for novice users. **Fit Criterion:** Advanced trading indicators and portfolio analytics shall remain collapsed or secondary until the user completes an initial trade or logs at least three days of physical activity. **Related Requirements:** UHR-EOU-08

ID	Requirement	Specification
TRN-03	Demonstrator Quick-Start Kit	**Description:** The team shall provide a demonstrator guide and automated demo-data seeding script for project evaluations and presentations. **Rationale:** Preloaded data enables evaluators to verify application features without completing real-time workouts. **Fit Criterion:** A preset configuration or debug toggle shall load 30 days of simulated activity and portfolio history within 10 seconds, enabling demonstrations of trading and compounding charts. **Related Requirements:** DOC-04

26 Waiting Room

The requirements below are stretch goals. They are out of scope for the capstone and will only be considered if time allows.

ID	Requirement	Specification
FUNC-17	Sponsored Reward Redemption	**Description:** The system shall allow users to redeem virtual currency for discounts, coupons, products, or services provided by participating sponsors. **Rationale:** Real-world rewards strengthen user engagement by connecting fitness achievements to tangible benefits. This is a stretch goal because it needs sponsor partnerships the team does not have, and because it gives the currency real-world value, which the Hazard Analysis assumes it does not have. **Fit Criterion:** Users shall be able to view available rewards, view redemption costs, and successfully redeem rewards when sufficient virtual currency is available. **Related Requirements:** COMP-LEG-03, SC-05

27 Ideas for Solution

N/A
Reason: To avoid focusing on the "how" within this requirements document, the team has not included any ideas for solution. The main purpose of this requirements document is to focus on the "what".

28 Appendix — Traceability Matrix

Table 7 lists, for every requirement and constraint, the requirements it is related to. A relationship appears in both rows: if A refers to B, or B refers to A, each is listed under the other. The same list appears in each requirement's *Related Requirements* field. All IDs are links. Table 8 maps each safety requirement in the Hazard Analysis to the SRS requirements that satisfy it.

Table 7: Traceability matrix

ID	Requirement	Related requirements
FUNC-01	User Registration	UHR-EOU-01, PR-SC-04, PR-SC-05, PROD-02, SEC-ACC-04, COMP-LEG-06
FUNC-02	User Authentication	UHR-EOU-04, SEC-ACC-01, SEC-ACC-02, SEC-ACC-04, SEC-ACC-05, SEC-IMM-02
FUNC-03	Third-Party Fitness Data Import	UHR-EOU-06, PR-SL-04, PR-PA-03, PR-PA-05, PR-RFT-01, PR-RFT-03, PR-SE-01, MAS-MNT-01, MAS-MNT-04, SEC-ACC-03, SEC-INT-01, COMP-LEG-01, COMP-LEG-03, DOC-02, SC-04
FUNC-04	Fitness Goal Creation	UHR-PER-01, UHR-LRN-03, UHR-ACC-06, PR-SC-06
FUNC-05	Fitness Goal Tracking	UHR-INT-01, PR-SC-02, PR-PA-05
FUNC-06	Reward Allocation	UHR-UND-03, PR-SC-01, PR-PA-02, PR-PA-03, SEC-INT-01, SEC-AUD-01, SEC-IMM-03
FUNC-07	Virtual Currency Tracking	LF-STY-02, PR-PA-01, PR-RFT-04
FUNC-08	Portfolio Creation	PR-RFT-02
FUNC-09	Asset Purchase	UHR-LRN-02, UHR-UND-03, PR-SL-03, PR-SC-03, PR-RFT-02, PR-SE-02, SEC-INT-02, SEC-IMM-02, SC-05
FUNC-10	Asset Sale	UHR-UND-03, PR-SL-03, PR-RFT-02, SEC-INT-02, SC-05
FUNC-11	Transaction Recording	PR-C-02, SEC-AUD-01

ID	Requirement	Related requirements
FUNC-12	Portfolio Performance Calculation	LF-APP-02, UHR-UND-02, PR-PA-04, PR-PA-06, PR-RFT-04
FUNC-13	Leaderboard Participation	UHR-PER-03, UHR-PER-04, UHR-POL-04, PR-SC-08, SEC-PRV-02, SEC-IMM-03
FUNC-14	Friend Management	UHR-PER-04, UHR-POL-05
FUNC-15	Private Leaderboards	UHR-PER-04, PR-SC-08
FUNC-16	Progress Reporting	LF-APP-01, UHR-EOU-02, PR-RFT-04, PR-C-02, PR-L-02
LF-APP-01	Visual Hierarchy	FUNC-16, UHR-EOU-02
LF-APP-02	Compounding Chart	FUNC-12, UHR-ACC-03
LF-APP-03	Responsive Layout	UHR-ACC-02, UHR-ACC-04, PR-L-01, PROD-02
LF-APP-04	Theme and Contrast	UHR-ACC-03, COMP-STD-02
LF-APP-05	Navigation	UHR-EOU-03
LF-STY-01	Approachable Tone	UHR-UND-01, UHR-POL-01
LF-STY-02	Virtual Currency Symbol	FUNC-07, PR-SC-03
LF-STY-03	Non-Intrusive Gamification	UHR-POL-02, PR-SL-02
LF-STY-04	Component Consistency	UHR-EOU-07
UHR-EOU-01	Onboarding Completion Time	FUNC-01, UHR-LRN-01
UHR-EOU-02	Dashboard Accessibility	FUNC-16, LF-APP-01
UHR-EOU-03	Minimal Screen Navigation	LF-APP-05
UHR-EOU-04	Error Feedback and Recovery	FUNC-02, MAS-SUP-01
UHR-EOU-05	Positive Feedback	UHR-UND-03, PR-SL-02, PR-SL-03
UHR-EOU-06	Simple and Reliable Device Connectivity	FUNC-03, DOC-02
UHR-EOU-07	Consistent User Interface	LF-STY-04
UHR-EOU-08	Learnability	UHR-EOU-09, UHR-LRN-01, TRN-01, TRN-02
UHR-EOU-09	Efficient Re-Learnability	UHR-EOU-08

ID	Requirement	Related requirements
UHR-PER-01	Personalized Fitness Goals	FUNC-04
UHR-PER-02	Customizable Notifications	UHR-ACC-05, PR-SC-07
UHR-PER-03	Customizable Privacy Settings	FUNC-13
UHR-PER-04	Customizable Competition Groups	FUNC-13, FUNC-14, FUNC-15
UHR-INT-01	Canadian Time Zone Support	FUNC-05, PR-PA-05, CUL-01
UHR-LRN-01	Minimal User Training	UHR-EOU-01, UHR-EOU-08, TRN-01
UHR-LRN-02	Learning Curve of Financial Concepts	FUNC-09, UHR-LRN-04, DOC-01
UHR-LRN-03	Learning Curve of Fitness Based Concepts	FUNC-04, UHR-LRN-04
UHR-LRN-04	Integrated Educational Assistance	UHR-LRN-02, UHR-LRN-03, UHR-UND-01, CUL-02, COMP-LEG-05, DOC-01
UHR-UND-01	Plain Language Communication	LF-STY-01, UHR-LRN-04, CUL-01, CUL-02
UHR-UND-02	Compounding Transparency and Distinction	FUNC-12
UHR-UND-03	Action Outcome Transparency	FUNC-06, FUNC-09, FUNC-10, UHR-EOU-05
UHR-POL-01	Respectful User Communication	LF-STY-01, UHR-POL-03, UHR-POL-04, CUL-02
UHR-POL-02	Positive Motivational Feedback	LF-STY-03, PR-SC-07
UHR-POL-03	Non-Punitive Performance Messaging	UHR-POL-01, PR-SC-07
UHR-POL-04	Respectful Competitive Features	FUNC-13, UHR-POL-01
UHR-POL-05	Respectful User-Generated Content	FUNC-14
UHR-ACC-01	Screen Reader Compatibility	COMP-STD-02

ID	Requirement	Related requirements
UHR-ACC-02	Scalable Text Support	LF-APP-03, COMP-STD-02
UHR-ACC-03	Color Independent Information	LF-APP-02, LF-APP-04, COMP-STD-02
UHR-ACC-04	Touch Target Accessibility	LF-APP-03, COMP-STD-02
UHR-ACC-05	Alternative Notification Methods	UHR-PER-02, COMP-STD-02
UHR-ACC-06	Inclusive Activity Tracking	FUNC-04, MAS-MNT-01
PR-SL-01	App launch	PR-C-01
PR-SL-02	Response to input	LF-STY-03, UHR-EOU-05, PR-C-01, PR-C-02, PR-SE-03
PR-SL-03	Trades	FUNC-09, FUNC-10, UHR-EOU-05, PR-C-01, PR-SE-03
PR-SL-04	Activity sync	FUNC-03
PR-SC-01	Daily earning limits	FUNC-06
PR-SC-02	No penalties for missed days	FUNC-05
PR-SC-03	Simulated market is labelled	FUNC-09, LF-STY-02
PR-SC-04	Not financial advice	FUNC-01
PR-SC-05	Exercise injury warning	FUNC-01
PR-SC-06	Safe goal progression	FUNC-04
PR-SC-07	Notification limits	UHR-PER-02, UHR-POL-02, UHR-POL-03
PR-SC-08	Safe leaderboard rules	FUNC-13, FUNC-15, SEC-PRV-02
PR-PA-01	Exact money values	FUNC-07
PR-PA-02	Consistent currency awards	FUNC-06
PR-PA-03	No double counting	FUNC-03, FUNC-06, SEC-INT-01
PR-PA-04	Portfolio value	FUNC-12
PR-PA-05	Workout dates	FUNC-03, FUNC-05, UHR-INT-01
PR-PA-06	Realistic market simulation	FUNC-12
PR-RFT-01	Offline activity	FUNC-03
PR-RFT-02	Atomic writes	FUNC-08, FUNC-09, FUNC-10
PR-RFT-03	Failing data sources	FUNC-03, MAS-MNT-04

ID	Requirement	Related requirements
PR-RFT-04	Backend unavailable	FUNC-07, FUNC-12, FUNC-16
PR-C-01	Users	PR-SL-01, PR-SL-02, PR-SL-03, PR-SE-03
PR-C-02	History	FUNC-11, FUNC-16, PR-SL-02
PR-SE-01	Adding fitness sources	FUNC-03, MAS-ADPT-01
PR-SE-02	Adding asset types	FUNC-09, MAS-MNT-02
PR-SE-03	More server capacity	PR-SL-02, PR-SL-03, PR-C-01
PR-L-01	Supported Android versions	LF-APP-03, PROD-02, COMP-STD-01
PR-L-02	Keeping user history	FUNC-16
PROD-01	Build and Quality Assurance	MAS-MNT-03, COMP-STD-06, SC-06
PROD-02	Installability	FUNC-01, LF-APP-03, PR-L-01, COMP-STD-01
MAS-MNT-01	Fitness Activity Addition Effort	FUNC-03, UHR-ACC-06
MAS-MNT-02	Investment Asset Addition Effort	PR-SE-02
MAS-MNT-03	Corrective Maintenance Effort	PROD-01, MAS-SUP-01
MAS-MNT-04	Third-Party Integration Maintenance	FUNC-03, PR-RFT-03, MAS-ADPT-01
MAS-MNT-05	Traceability	COMP-LEG-07, DOC-04
MAS-SUP-01	Contact Support Capability	UHR-EOU-04, MAS-MNT-03, DOC-03
MAS-ADPT-01	External Fitness Provider Adaptability	PR-SE-01, MAS-MNT-04, DOC-02
MAS-ADPT-02	Mobile Platform Portability	SC-01
MAS-ADPT-03	Future Artificial Intelligence Service Adaptability	COMP-LEG-05
SEC-ACC-01	Authenticated Access	FUNC-02, COMP-LEG-02, COMP-STD-07
SEC-ACC-02	Role-Based Authorization	FUNC-02, SEC-PRV-02

ID	Requirement	Related requirements
SEC-ACC-03	Fitness Integration Authorization	FUNC-03
SEC-ACC-04	Password and Sign-In Protection	FUNC-01, FUNC-02, SEC-AUD-02, SEC-IMM-02
SEC-ACC-05	Session End and Device Unlinking	FUNC-02
SEC-INT-01	Activity Validation	FUNC-03, FUNC-06, PR-PA-03, SEC-AUD-02, SEC-IMM-03
SEC-INT-02	Server-Side Valuation	FUNC-09, FUNC-10
SEC-INT-03	Data Backups and Recovery	SEC-PRV-03
SEC-PRV-01	Health Data Protection	COMP-LEG-02, COMP-STD-07
SEC-PRV-02	Private by Default	FUNC-13, PR-SC-08, SEC-ACC-02, COMP-LEG-02
SEC-PRV-03	User Data Deletion	SEC-INT-03, SEC-AUD-01, COMP-LEG-04
SEC-AUD-01	Economic Event Logging	FUNC-06, FUNC-11, SEC-PRV-03
SEC-AUD-02	Security Event Logging	SEC-ACC-04, SEC-INT-01, SEC-IMM-02
SEC-IMM-01	Injection Defense	COMP-STD-07
SEC-IMM-02	Rate Limiting	FUNC-02, FUNC-09, SEC-ACC-04, SEC-AUD-02
SEC-IMM-03	Client Environment Validation	FUNC-06, FUNC-13, SEC-INT-01
CUL-01	Supports Canadian Languages	UHR-INT-01, UHR-UND-01
CUL-02	Culturally Neutral Educational Content	UHR-LRN-04, UHR-UND-01, UHR-POL-01
COMP-LEG-01	User Consent for Health Data Collection	FUNC-03
COMP-LEG-02	Protection of Personal Information	SEC-ACC-01, SEC-PRV-01, SEC-PRV-02, COMP-LEG-04
COMP-LEG-03	Sharing of User Data	FUNC-03, FUNC-17
COMP-LEG-04	User Data Deletion Rights	SEC-PRV-03, COMP-LEG-02
COMP-LEG-05	AI-Content Disclosure	UHR-LRN-04, MAS-ADPT-03

ID	Requirement	Related requirements
COMP-LEG-06	Age Verification	FUNC-01
COMP-LEG-07	Third-Party Licensing Compliance	MAS-MNT-05, SC-02, SC-03, SC-04
COMP-STD-01	Android Platform Compliance	PR-L-01, PROD-02, SC-06
COMP-STD-02	Accessibility Compliance	LF-APP-04, UHR-ACC-01, UHR-ACC-02, UHR-ACC-03, UHR-ACC-04, UHR-ACC-05
COMP-STD-03	TypeScript Coding Standards Compliance	SC-02
COMP-STD-04	Java Coding Standards Compliance	SC-03
COMP-STD-05	Database Compliance	SC-03
COMP-STD-06	Software Testing Standard Compliance	PROD-01
COMP-STD-07	Information Security Compliance	SEC-ACC-01, SEC-PRV-01, SEC-IMM-01
DOC-01	Financial and Fitness Glossary	UHR-LRN-02, UHR-LRN-04
DOC-02	Fitness Integration Guide	FUNC-03, UHR-EOU-06, MAS-ADPT-01
DOC-03	Searchable Help Center	MAS-SUP-01
DOC-04	Digital Manual and Release Notes	MAS-MNT-05, TRN-03
TRN-01	Interactive Onboarding	UHR-EOU-08, UHR-LRN-01
TRN-02	Progressive Feature Discovery	UHR-EOU-08
TRN-03	Demonstrator Quick-Start Kit	DOC-04
FUNC-17	Sponsored Reward Redemption	COMP-LEG-03, SC-05
SC-01	Android platform	MAS-ADPT-02
SC-02	Frontend technology	COMP-LEG-07, COMP-STD-03
SC-03	Backend technology	COMP-LEG-07, COMP-STD-04, COMP-STD-05
SC-04	Fitness data sources	FUNC-03, COMP-LEG-07

ID	Requirement	Related requirements
SC-05	Simulated market only	FUNC-09, FUNC-10, FUNC-17
SC-06	Distribution	PROD-01, COMP-STD-01

Table 8: Hazard Analysis safety requirements and the SRS requirements that satisfy them

Safety requirement	SRS requirements
SR1	SEC-ACC-04
SR2	SEC-ACC-05
SR3	SEC-ACC-02, SEC-PRV-02
SR4	UHR-EOU-04, PR-RFT-03
SR5	PR-PA-05
SR6	PR-SC-06
SR7	PR-SC-07
SR8	PR-SC-01, PR-SC-02
SR9	PR-SC-08, SEC-PRV-02
SR10	PR-PA-06
SR11	PR-SC-03, PR-SC-04
SR12	PR-RFT-02, SEC-INT-03

29 Appendix — Generative AI Use Disclosure

29.1 AI Tools Used

- **Gemini 3.8**
- **ChatGPT 5.6**
- **GitHub Copilot**, using the Generative Pre-trained Transformer 5 (GPT-5) model
- **Claude Opus 5.5**
- **Claude Opus 5.5**

29.2 How and Where Used

The AI tools listed above were employed across several phases of drafting and refining the Software Requirements Specification:

- **Ideation and Conceptual Brainstorming:** Generative AI tools were used during initial requirement gathering and brainstorming sessions to explore feature ideas, gamification mechanics, and educational workflows for connecting physical fitness to financial compounding.
- **Persona Creation (Section 2):** Generative AI tools were used to generate realistic, representative personas (Brandon, Connor, and Daniella) using curated prompts. These prompts guided the models to simulate distinct user archetypes with realistic backgrounds, habits, motivations, and pain points relevant to the application's target demographics.
- **Business Model Diagramming (Section 7):** AI tools were utilized to convert initial conceptual sketches of the business model and data architecture into structured PlantUML code, which was subsequently rendered into formal system diagrams.
- **Document Review and Clarity Refinement (Document-wide):** The AI tools were used to review draft sections for consistency, phrasing, and grammatical flow. Content was reworded where necessary to

eliminate ambiguity, enhance readability, and adhere to formal technical documentation standards.

- **Section Drafting and Consistency Review (Sections 3, 12, and 19):** Claude was used to draft the Mandated Constraints, Performance Requirements, and Off-the-Shelf Solutions sections from the team's Problem Statement, Development Plan, and Hazard Analysis. It was also used to review the whole document for inconsistencies between sections and against the course's SRS rubric, and to make the resulting changes: merging duplicated requirements, rewording requirements to be verifiable and free of design decisions, and adding the conventions, symbolic constants, glossary, likely and unlikely changes, and traceability matrix. Team members made every scope decision used to resolve the inconsistencies, and reviewed and edited all of the changes before they were merged.

- **LaTeX Formatting, Images, and Tables (Document-wide):** The tools assisted in authoring and debugging LaTeX markup, particularly the configuration, alignment, and formatting of multi-page requirement tables (`longtable` and `xltabular`) as well as figure layout environments.

29.3 Verification of AI-Generated Content

All content produced or modified with AI assistance was verified by the team to ensure accuracy, technical correctness, and alignment with project goals.

- **Peer Review and Cross-Reading:** Every section developed or refined using AI was reviewed by multiple team members through peer reviews and pull request evaluations to ensure the content accurately reflected project goals and engineering specifications.

- **Source and Fact Verification:** All factual assertions, financial principles, domain definitions, and references were verified against authoritative academic sources, financial standards, and official platform documentation to guard against hallucinated or fabricated claims.

- **Diagram and Implementation Validation:** The PlantUML code, architectural models, and state diagrams generated with AI assistance

were validated by the team against our planned backend data schema and system requirements to ensure technical correctness and consistency.

- **Document and Compilation Testing:** All LaTeX markup, tables, and environments generated with AI assistance were compiled locally to ensure syntax validity, proper page flow, and adherence to document formatting standards.

The team assumes full responsibility for the accuracy, completeness, consistency, and appropriateness of this document.

29.4 User Interview

Interview Notes (Rough Summary)

Initial Spending Question

Q: If given $1000 to spend freely, what would you do with it?
A: Would contribute the money to savings; demonstrates a savings-oriented mindset.

Financial Literacy (Self-Rating)

Rating: 2 out of 5.
Details: Understands general budgeting and good spending habits. Limited knowledge of investment-specific topics. No strong personal interest in finance, but acknowledges the importance of learning more. Lack of interest and competing priorities reduce motivation.

Personal Fitness (Self-Rating)

Rating: 3 out of 5.
Details: Works out approximately three times per week. Maintains clean, balanced, and trackable eating habits. Does not use supplements. Fitness goals are broad and qualitative (e.g., "get stronger, feel better") rather than specifically tracked.

Current Tools and Devices

Fitness Devices: Apple Watch (primary tracker for workouts, steps, calories, heart rate). Previously used an Ultrahuman Ring.
Financial Tools: Relies mainly on bank advisors; no mention of dedicated apps or platforms.

Intimidating or Confusing Topics

Finance: Anything beyond basic concepts feels uninteresting or overly complex. Investment topics beyond surface level are intimidating. Relies on bank advisors for guidance.
Fitness: Lifestyle does not feel intimidating; good habits maintained through personal motivation.

Motivation and Confidence Factors

Finance: Lack of interest is the main barrier. Would benefit from structured, guided learning. Prefers simple, objective-based tasks.
Fitness: Already confident; motivation comes from personal drive rather than external tools.

Habit Formation (Combined Score Below 7)

Finance Habits: Basics upheld through budgeting; deeper learning hindered by low interest.
Fitness Habits: Maintained through personal motivation and routine.

Additional Insights

Interest Drivers: Daily or objective-based games increase engagement (e.g., Wordle). Prefers one clear objective per day with a guided approach. Fitness is already a stable part of lifestyle; finance would provide more value if gamified or structured.

30 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. **What went well while writing this deliverable?**

   Overall, the team maintained a high level of efficiency and productivity throughout the completion of this deliverable. Following our planning meetings, each team member took full accountability and stayed on top of their assigned tasks and sections, ensuring continuous forward momentum.

   We established strong alignment across the entire team in terms of collaboration expectations, communication cadence, and work quality. When technical or domain challenges arose, such as diagnosing and resolving Git merge issues or standardizing the formal phrasing and fit criteria for Volere requirements—the team collaborated effectively through ad-hoc calls and discussions to solve problems collectively.

   Additionally, the team successfully adapted to significant variations in individual availability and geographic dispersion. Despite several members taking vacations or traveling across distinct time zones (specifically with Noah traveling in Europe and Arjun located in Vancouver), the group maintained continuous communication through Microsoft Teams and GitHub. We adapted through asynchronous workflows, flexible scheduling, and clear issue tracking, ensuring that all sections progressed without disruption.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   The team experienced several key pain points during this deliverable cycle, which were addressed through proactive adjustments and teamwork:

   - **Disparate Availability During Reading Week:** A large portion of the work occurred over Reading Week, during which team members experienced starkly differing availability schedules, vacations, and significant time-zone differences (Noah traveling in Europe and Arjun located in Vancouver). Finding mutual times for synchronous meetings proved difficult. We resolved this by relying primarily on asynchronous updates in Microsoft Teams, breaking work into modular GitHub issues, and holding focused, flexible meetings when schedules overlapped.
   - **Git PR Overload and Branch Protection Bottlenecks:** Because team members worked concurrently across multiple feature branches to draft different sections of the extensive Volere document, a surge of Pull Requests arrived simultaneously right before the deadline. This led to Git merge conflicts, review backlogs, and pipeline friction. To resolve this under tight time constraints, we temporarily disabled GitHub branch protection rules and joined a live group call. We reviewed, tested, and merged all branches together in real time, verifying document integrity and preventing regressions.
   - **Rushed Development Plan Integration:** Because branch merges took place until the day before the deadline, the downstream sections—notably the development plan and schedule—had to be completed last. Consequently, completing and polishing this section felt somewhat rushed. We resolved this by rallying as a group during our review call to finalize the section collaboratively, and we established an internal policy to enforce earlier merge freezes for future deliverables.
   - **Estimating Performance Requirements Without Baselines:** Formulating quantitative, measurable performance requirements was challenging because we were not provided with baseline benchmarks or existing system metrics. Establishing realistic thresholds

for latency, throughput, and synchronization required significant estimation. We resolved this by conducting a dedicated team walkthrough, researching performance benchmarks of comparable commercial fitness and personal finance applications, and collaboratively establishing reasonable, verifiable metrics.

3. **How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?**

   A substantial portion of our requirements—particularly our functional requirements—were directly inspired by stakeholder and client discussions. The team conducted two primary interviews:

   (a) A formal mock interview with our client and supervisor, Chris Schankula, which helped establish the fundamental project scope, define educational learning goals, and clarify expectations around combining fitness habit-tracking with financial simulations.

   (b) A dedicated user interview with a representative end-user (documented in the Appendix), which provided direct insight into real-world fitness routines, financial literacy levels, and motivational drivers.

   These interviews revealed that while fitness habits are often self-motivated through routine, financial education requires structured, bite-sized gamification (such as daily objective challenges inspired by Wordle) to maintain user engagement. These insights directly inspired functional requirements regarding daily interactive challenges, simplified financial glossaries, and automated tracking integrations. Because our team has chosen to produce a comprehensive User Interview report as our Custom Document deliverable, these early interviews were pivotal in grounding our requirements in genuine user and client needs.

4. **Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.**

   A cross-disciplinary set of software engineering and commerce courses provides the theoretical and practical knowledge needed for our capstone project:

- **SFWRENG 3RA3 (Software Requirements and Security Considerations):** Provided the direct framework for requirement engineering, writing unambiguous functional and non-functional requirements with the Volere template, establishing fit criteria, maintaining requirements traceability, and incorporating critical security and privacy considerations directly into our system requirements.
- **SFWRENG 2AA4 (Software Design I — Introduction to Software Development):** Taught modular decomposition, information hiding, object-oriented principles, design patterns, automated unit testing, and disciplined version control workflows with Git.
- **SFWRENG 3A04 (Software Design III — Large System Design):** Provided experience in high-level architectural styles (e.g., client-server, layered architecture), statechart and UML modeling, formal interface specifications, and decomposing complex software applications into manageable subsystems.
- **COMMERCE 2FA3 (Introduction to Finance):** Introduced core personal and corporate finance concepts, the time value of money, compounding interest, financial market mechanisms, equity and fixed-income valuation, and risk/return principles that form the basis of the app's financial simulation engine.
- **COMMERCE 2FB3 (Managerial Finance):** Built upon financial fundamentals with capital budgeting, financial valuation, and cash flow analysis, informing the design of realistic wealth-growth projections, budgeting logic, and portfolio tracking features.
- **SFWRENG 3DB3 (Databases):** Provided essential skills in relational database modeling, entity-relationship (ER) diagrams, normalization (1NF–3NF), SQL database design, transaction management, and concurrency control, which will guide the architecture of our user data schema and persistence layer.
- **SFWRENG 4X03 (Scientific Computation):** Provided knowledge of numerical methods, computer arithmetic error analysis, and mathematical algorithms, ensuring numerical precision and

stability when executing financial compounding models, interest simulations, and fitness metric calculations.

5. **What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project?**

   To successfully execute and complete the Flexin capstone project, each team member has identified a specific domain knowledge area or technical skill they collectively need to acquire:

   - **Daniel Hu — Financial Domain Knowledge:** Because Daniel H. is enrolled in the core Software Engineering stream rather than Software Engineering and Management, he needs to acquire domain-specific knowledge in personal finance, financial instruments, compounding mechanics, portfolio diversification, and investment valuation to ensure the app's financial calculations and educational simulations are mathematically sound and realistic.
   - **Noah Goldschmied — Frontend Development:** Noah needs to acquire practical skills in modern mobile frontend development frameworks (e.g., React Native/Expo or Flutter), specifically mastering component lifecycle management, mobile user interface design, styling, and navigation routing.
   - **Daniel Franze-Da Silva — Backend Systems Development:** Daniel F. needs to acquire advanced backend engineering capabilities, focusing on architecting scalable server-side systems, RESTful API design, token-based authentication/authorization (JWT/OAuth), and robust business logic processing.
   - **Peter Deforest — Database Design:** Peter needs to acquire specialized knowledge in database schema architecture, relational database normalization (3NF), data integrity constraints, migrations, and query optimization (using PostgreSQL or SQLite) to support efficient data storage for fitness telemetry and financial history.
   - **Arjun Karthik — Frontend System Development:** Arjun needs to acquire deep expertise in frontend system architecture, specifically client-side state management patterns (such as Redux Toolkit or Zustand), asynchronous data synchronization with backend APIs, client caching, and offline state handling.

6. **For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?**

   For each of the knowledge areas and skills identified in Question 5, the team evaluated two distinct learning approaches and selected the path best suited to project success:

   - **Daniel Hu — Financial Domain Knowledge:**
     - *Approach 1 (Self-Directed Study via Literature and Online Guides):* Reading financial textbooks, verified educational platforms (e.g., Investopedia), and personal finance literature independently.
     - *Approach 2 (Peer-Mentoring and Collaborative Knowledge Transfer):* Participating in structured peer-learning sessions with team members enrolled in the Engineering and Management stream who have taken COMMERCE 2FA3 and 2FB3, reviewing relevant lecture slides and financial model templates.
     - *Choice and Justification:* Daniel will pursue **Approach 2** supplemented by targeted reading from Approach 1. Peer-mentoring allows Daniel to focus specifically on the financial concepts, formulas, and market mechanics that directly apply to Flexin's features, ensuring consistency across the team while avoiding extraneous theoretical topics outside the project scope.
   - **Noah Goldschmied — Frontend Development:**
     - *Approach 1 (Online Coursework and Documentation):* Completing guided video tutorials (e.g., React Native/Flutter courses) and systematically working through official framework documentation.
     - *Approach 2 (Rapid Interactive Prototyping):* Developing experimental sandboxed UI components, interactive wireframes, and proof-of-concept mobile screens for Flexin.
     - *Choice and Justification:* Noah will pursue **Approach 2**. Rapid prototyping provides immediate practical feedback, ac-

celerates mastery of layout styling and component composition, and directly produces reusable UI screens that contribute to project deliverables.

- **Daniel Franze-Da Silva — Backend Systems Development:**
  - *Approach 1 (Theoretical Architectural Study):* Reading backend engineering literature and API design patterns focusing on microservices, request pipelining, and system security.
  - *Approach 2 (Iterative API Prototyping and Testing):* Building functional mock backend endpoints, configuring routing and middleware, and performing live testing using API tools such as Postman.
  - *Choice and Justification:* Daniel will pursue **Approach 2**. Iterative prototyping allows him to encounter and resolve concrete challenges in endpoint structure, JSON serialization, and status-code handling in real time, building practical backend competence that directly benefits the application.
- **Peter Deforest — Database Design:**
  - *Approach 1 (Textbook Study and Query Optimization Guides):* Studying relational database theory, database normalization rules, and query execution plans from textbooks and online documentation.
  - *Approach 2 (Iterative Schema Construction and Stress Testing):* Creating mock relational database schemas (e.g., SQLite/PostgreSQL), writing schema migration scripts, and executing realistic query workloads against simulated fitness and transaction data.
  - *Choice and Justification:* Peter will pursue **Approach 2** alongside reference material from Approach 1. Building and executing actual schema migrations against test data allows Peter to identify data integrity issues and relational constraints early, ensuring that the database schema is production-ready.
- **Arjun Karthik — Frontend System Development:**
  - *Approach 1 (Open-Source Codebase Analysis):* Studying production-grade open-source mobile applications to observe how global state stores, middleware, and caching layers are organized.

- *Approach 2 (Isolated Integration Spikes):* Building isolated test benches to connect UI components with mock API endpoints, focusing on handling loading states, cache invalidation, and asynchronous error boundaries.
- *Choice and Justification:* Arjun will pursue **Approach 1** combined with practical integration spikes from Approach 2. Analyzing established, mature open-source repositories provides immediate insight into proven architectural patterns, helping avoid state-management anti-patterns early in system design.